

A Survey of Large Language Models for Generative Recommendation and User Behavior Modeling

InteractiveSurvey

Abstract

The rapid evolution of digital content has driven a paradigm shift in recommender systems, transitioning from traditional collaborative filtering to sophisticated architectures powered by Large Language Models (LLMs). This survey critically examines the transformative role of LLMs in redefining recommendation as a generative task, encompassing unified frameworks, retrieval-augmented generation, and agent-based reasoning. We propose a novel taxonomy categorizing existing literature by generative paradigms and knowledge integration mechanisms, while analyzing how semantic identifier tokenization and joint search-recommendation modeling bridge the gap between collaborative signals and rich semantic contexts. Furthermore, the study synthesizes strategies for mitigating hallucinations through structured knowledge constraints and explores dynamic context injection techniques that enable systems to perform sophisticated reasoning over evolving user preferences without rigid state tracking. By evaluating cognitive architectures that facilitate interactive, narrative-driven explanations, this work highlights the transition from passive filtering to active, reasoning partnerships in user decision-making. Ultimately, this comprehensive analysis outlines a strategic roadmap for developing next-generation recommender systems that are not only accurate but also explainable, adaptive, and deeply aligned with human intent.

1 Introduction

The rapid proliferation of digital content has necessitated a fundamental evolution in recommender systems, shifting from traditional collaborative filtering to sophisticated deep learning architectures capable of handling massive, heterogeneous data. While conventional models excel at capturing explicit user-item interactions through matrix factorization or graph neural networks, they

often struggle with semantic understanding, cold-start scenarios, and the integration of unstructured contextual information. The emergence of Large Language Models (LLMs) has catalyzed a paradigm shift, offering unprecedented capabilities in natural language understanding, reasoning, and generative synthesis [1]. By leveraging vast pre-trained knowledge and flexible tokenization schemes, LLMs enable recommenders to transcend simple pattern matching, facilitating a deeper comprehension of user intent and item semantics within a unified generative framework [2].

This survey paper critically examines the transformative role of Large Language Models in generative recommendation and user behavior modeling [3]. Unlike previous reviews that focus primarily on discriminative approaches or isolated applications of language models, this work provides a comprehensive synthesis of methodologies where recommendation is framed as a generative task [4]. We explore how LLMs redefine the core components of recommender systems, from item representation and retrieval mechanisms to interactive reasoning and explanation generation [5]. The central thesis posits that integrating LLMs allows for a seamless unification of search, recommendation, and conversational interaction, effectively bridging the gap between collaborative signals and rich semantic contexts to create more adaptive, interpretable, and robust systems [6].

The core of this survey begins by dissecting the architectural foundations of generative recommendation, specifically focusing on unified paradigms and identifier tokenization strategies [7]. We analyze how semantic IDs constructed via residual quantization replace opaque indices with structured token sequences, enabling models to share statistical strength across semantically similar items while mitigating vocabulary explosion. Furthermore, we investigate the joint modeling of search and recommendation tasks, where distinct

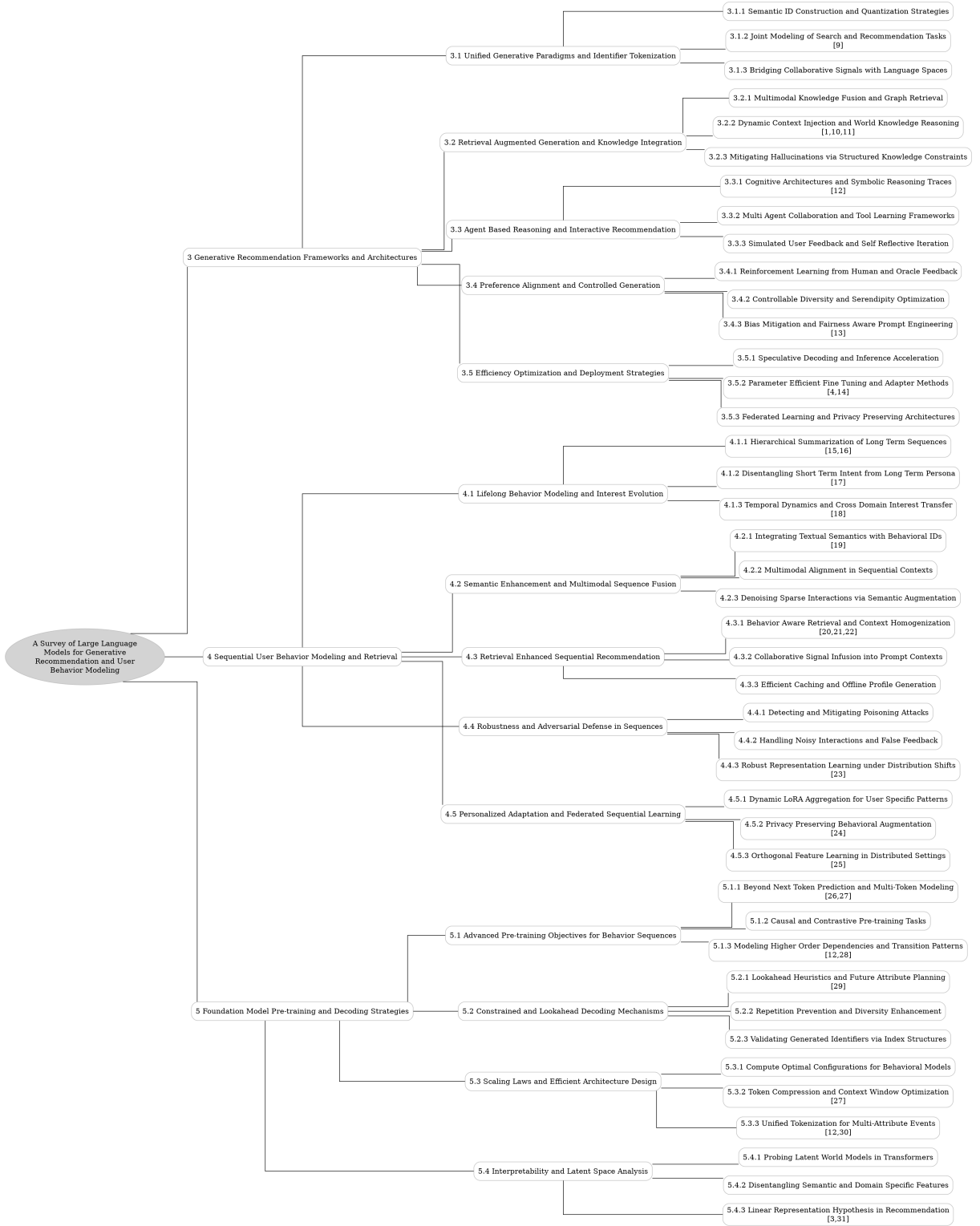


Figure 1: The outline of our survey: A Survey of Large Language Models for Generative Recommendation and User Behavior Modeling

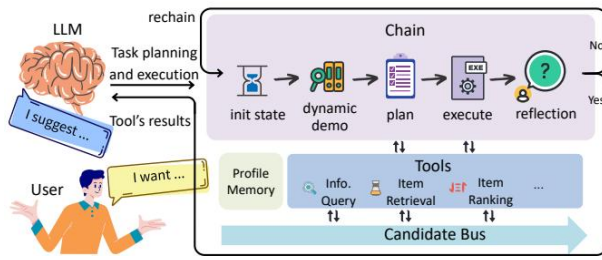


Figure 2: Chart from 'RecAI: Leveraging Large Language Models for Next-Generation Recommender Systems' (arXiv:2403.06465)

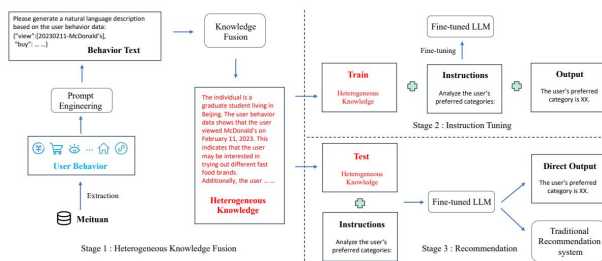


Figure 3: Chart from 'Heterogeneous Knowledge Fusion: A Novel Approach for Personalized Recommendation via LLM' (arXiv:2308.03333)

$$C(u) = \bigcup_{k=1}^K \text{Top}(S_k(u, i), n_k)$$

Figure 4: Chart from 'The Application of Large Language Models in Recommendation Systems' (arXiv:2501.02178)

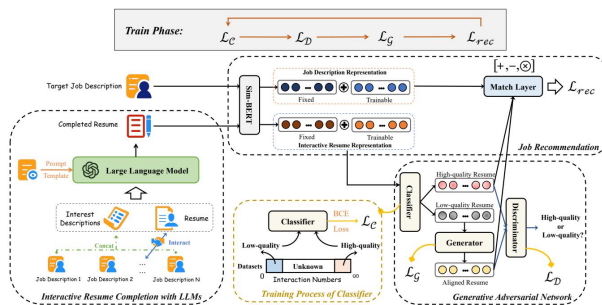


Figure 5: Chart from 'Enhancing Job Recommendation through LLM-based Generative Adversarial Networks' (arXiv:2307.10747)

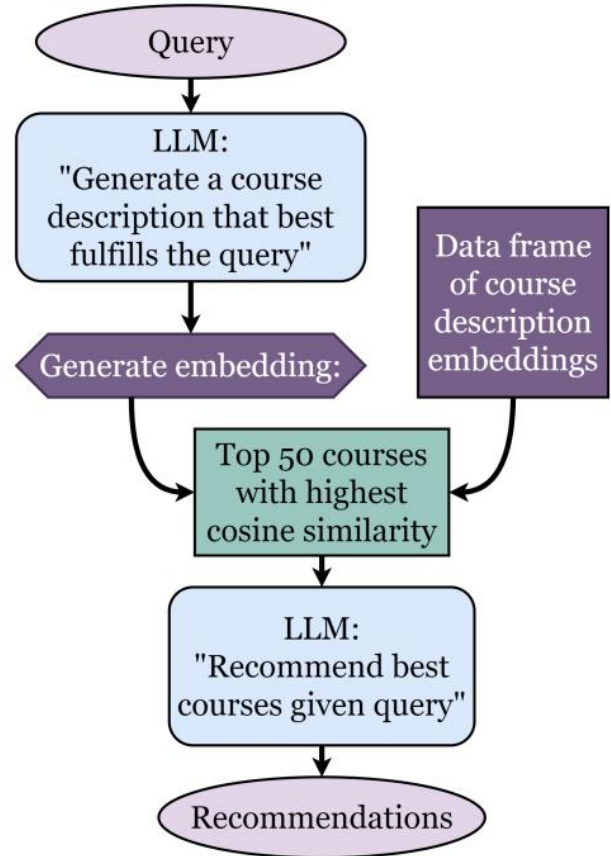


Figure 6: Chart from 'FROM INTERESTS TO INSIGHTS: AN LLM APPROACH TO COURSE RECOMMENDATIONS USING NATURAL LANGUAGE QUERIES' (arXiv:2412.19312)

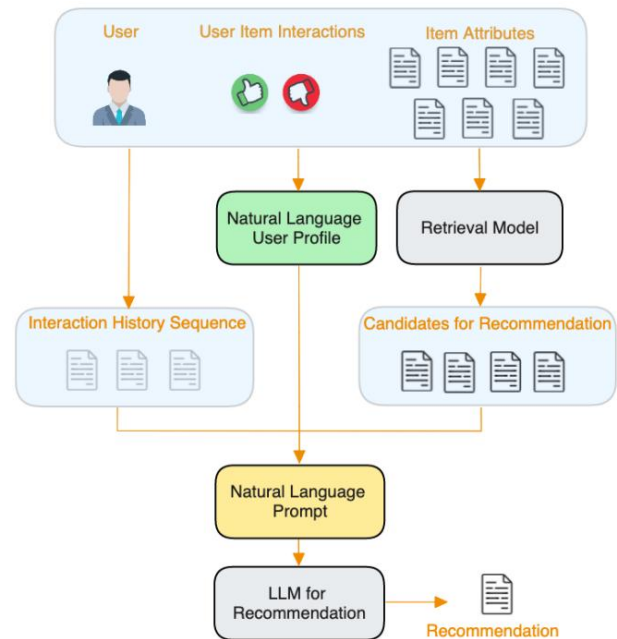


Figure 7: Chart from 'PALR: Personalization Aware LLMs for Recommendation' (arXiv:2305.07622)

silos are dissolved into a single generative framework that aligns explicit queries with implicit interaction histories. This section also addresses the critical challenge of bridging collaborative signals with language spaces, detailing hybrid architectures that inject graph-based relational knowledge directly into LLMs to preserve high-order connectivity alongside textual semantics.

$$\mathcal{L}_{\text{ranking}} = \frac{1}{|\mathcal{D}|} \sum_{(u,i) \in \mathcal{D}} (y_{ui} - \hat{r}_{ui})^2.$$

Figure 8: Chart from 'Legommenders : A Comprehensive Content-Based Recommendation Library with LLM Support' (arXiv:2412.15973)

Subsequently, the discussion advances to retrieval-augmented generation and the integration of external knowledge to enhance system reliability and scope. We examine multimodal knowledge fusion techniques that unify textual, visual, and structured metadata into shared semantic spaces, supported by graph-based retrieval mechanisms for efficient candidate selection. A significant portion is dedicated to dynamic context injection, illustrating how LLMs leverage inherent world knowledge to perform sophisticated reasoning over evolving user preferences without rigid state tracking. Crucially, we address the pervasive issue of hallucinations in generative systems, reviewing structured knowledge constraints and constrained decoding strategies that ground predictions in factual item catalogs, thereby ensuring the validity and trustworthiness of generated recommendations.

Finally, the survey explores the frontier of agent-based reasoning and interactive recommendation, marking a transition from static prediction to dynamic cognitive engagement. We evaluate cognitive architectures that utilize symbolic reasoning traces and instruction tuning to synthesize complex user interests from unstructured data, moving beyond limited slot-filling approaches. This section highlights how generative agents facilitate multi-turn dialogues and provide narrative-driven explanations that mirror human cognitive processes. By treating recommendation as an interactive reasoning problem, these systems achieve a level of personalization and adaptability previously unattainable, effectively transforming the recommender from a passive filter into an active,

reasoning partner in the user's decision-making journey.

The primary contributions of this survey are threefold [8]. First, we propose a novel taxonomy that categorizes existing literature based on generative paradigms, knowledge integration mechanisms, and reasoning architectures, providing a structured lens through which to view this rapidly expanding field. Second, we offer a critical analysis of the trade-offs between generation flexibility and factual grounding, synthesizing best practices for mitigating hallucinations while maximizing semantic richness. Third, by identifying current limitations and unresolved challenges, we outline a comprehensive roadmap for future research, guiding the development of next-generation recommender systems that are not only accurate but also explainable, interactive, and deeply aligned with human intent.

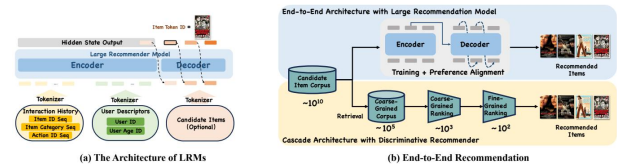


Figure 9: Chart from 'A Survey on Generative Recommendation: Data, Model, and Tasks' (arXiv:2510.27157)

2 Generative Recommendation Frameworks and Architectures

2.1 Unified Generative Paradigms and Identifier Tokenization

2.1.1 Semantic ID Construction and Quantization Strategies

Semantic ID construction fundamentally shifts item representation from discrete, opaque indices to structured token sequences that encode latent semantic relationships. Unlike traditional one-hot encodings, these identifiers are typically generated via Residual Quantized Variational Autoencoders (RQ-VAE), which map high-dimensional item content embeddings into hierarchical codebooks. This process decomposes continuous representations into a sequence of discrete tokens, where earlier tokens capture coarse-grained categories and subsequent tokens refine specific attributes, enabling the model to share statistical strength across semantically similar items while maintaining distinctiveness.

Quantization strategies play a pivotal role in balancing vocabulary efficiency with representation fidelity within this framework. By employing multi-stage residual quantization, methods can compress extensive item catalogs into manageable token sequences without incurring the vocabulary explosion associated with assigning unique tokens to every entity. Optimized product quantization techniques further enhance this by generating long, unordered semantic IDs in parallel, effectively integrating sub-item semantics. This hierarchical structure not only reduces the computational burden during autoregressive generation but also facilitates better generalization for cold-start items by leveraging shared semantic prefixes derived from content metadata.

Advanced inference mechanisms have been developed to address the challenges of generating valid semantic IDs at scale. Techniques such as Multi-Token Prediction (MTP) allow the simultaneous generation of multiple identifier tokens, significantly accelerating the recall phase compared to standard sequential decoding. Furthermore, replacing rigid Top-K selection with probabilistic sampling and matrix decomposition helps mitigate the risk of generating invalid or hallucinated identifiers that do not map to existing catalog items. These strategies collectively ensure that the generated token sequences remain grounded in the item space while maximizing the efficiency of large language model-based recommendation systems.

2.1.2 Joint Modeling of Search and Recommendation Tasks

Joint modeling of search and recommendation tasks represents a paradigm shift from treating these as distinct silos to unifying them under a single generative framework. Traditional approaches often rely on separate architectures for explicit query-based retrieval and implicit interest inference, leading to fragmented user representations. Recent advancements leverage Large Language Models (LLMs) to bridge this gap by aligning the input spaces of search queries and interaction histories [9]. By processing both explicit instructions and sequential behavior data through a shared transformer backbone, these models capture nuanced user intents that span immediate information needs and long-term preferences, effectively dissolving the boundary between active searching and passive recommendation.

The core mechanism of joint modeling involves

$$\mathcal{L}_{aug} = - \sum_{(u,pos,neg)} \ln \sigma(\hat{y}_{u,pos} - \hat{y}_{u,neg}),$$

Figure 10: Chart from 'Large Language Models as Data Augmenters for Cold-Start Item Recommendation' (arXiv:2402.11724)

transforming diverse interaction signals into a unified textual or embedding space where search and recommendation objectives are optimized simultaneously. In this setting, search queries are treated as explicit intent indicators, while historical clicks serve as implicit feedback, allowing the model to perform multi-task learning without specialized structural designs. Techniques such as instruction tuning enable the system to dynamically switch between retrieval modes based on context, utilizing autoregressive generation to predict future items or rank candidates. This approach facilitates the handling of complex scenarios, including personalized search and multi-objective optimization, by leveraging the LLM's inherent reasoning capabilities to interpret heterogeneous data sources cohesively.

Despite the theoretical advantages, practical implementation faces challenges regarding computational efficiency and the alignment of distinct optimization goals. Joint models must balance the precision required for query-specific retrieval with the diversity needed for serendipitous recommendations, often necessitating two-stage frameworks where efficient retrievers shortlist candidates for fine-grained LLM ranking. Furthermore, ensuring that the model retains performance on individual tasks while benefiting from shared representations remains critical. Current research focuses on developing robust grounding paradigms and reward mechanisms that guide the generative process, ensuring that the unified model delivers accurate, context-aware results across both search and recommendation domains without compromising scalability.

2.1.3 Bridging Collaborative Signals with Language Spaces

Bridging collaborative signals with language spaces addresses the fundamental semantic gap between discrete item identifiers used in traditional collaborative filtering and the continuous textual embeddings of large language models. Early approaches attempted to map collaborative seman-

tics directly into natural language prompts, utilizing templates or generated profiles to describe user interaction histories. However, these text-only representations often fail to capture high-order connectivity and latent relational patterns inherent in user-item graphs, limiting the model’s ability to perform inductive reasoning on unseen nodes or sparse interactions.

To overcome these limitations, recent methodologies focus on injecting structured collaborative information directly into the language model’s architecture. Techniques such as text-like encoding transform collaborative embeddings into token sequences that align with the LLM’s vocabulary, allowing the model to process relational knowledge alongside semantic content. Alternatively, hybrid architectures integrate ID embeddings as additional input features or modify the attention mechanisms to jointly attend to both textual descriptions and collaborative vectors, effectively unifying content-based semantics with behavioral signals within a single representation space.

Furthermore, advanced frameworks leverage graph neural networks to extract high-order collaborative signals before projecting them into the language domain, ensuring that complex user intent dynamics are preserved during fine-tuning. By aligning the training objectives of collaborative filtering with language modeling tasks, these approaches enable the system to interpret multi-turn dialogue contexts and session-based sequences more accurately. This integration not only enhances recommendation precision but also empowers the model to generate coherent, context-aware explanations that reflect both user preferences and global interaction trends.

2.2 Retrieval Augmented Generation and Knowledge Integration

2.2.1 Multimodal Knowledge Fusion and Graph Retrieval

Recent advancements in multimodal knowledge fusion leverage Large Language Models (LLMs) to unify heterogeneous data sources, including textual descriptions, visual content, and structured metadata, into shared semantic spaces. Unlike traditional late-fusion architectures that aggregate fixed embeddings, modern approaches employ end-to-end formulations where vision-language models align modality-specific structures through contrastive learning. This integration enables the

model to capture intricate cross-modal correlations, transforming disparate signals such as user behavior logs and item images into cohesive representations that significantly enhance personalization accuracy and interpretability.

To address the scalability challenges inherent in processing massive corpora, graph-based retrieval mechanisms have emerged as a critical component for efficient candidate selection. These systems construct knowledge graphs that encode complex user-item interactions and semantic relationships, facilitating multi-aspect retrieval via query decomposition. By utilizing graph neural networks, the retrieval phase effectively navigates high-dimensional spaces to filter irrelevant candidates, ensuring that only the most contextually pertinent items are passed to downstream ranking modules. This graph-driven approach mitigates information loss and hallucination by grounding generated recommendations in verified structural knowledge.

The synergy between multimodal fusion and graph retrieval culminates in sophisticated frameworks capable of dynamic reasoning over diverse data modalities. Techniques such as triple context restoration and query-driven feedback loops further refine the retrieval process, allowing the system to adaptively restore missing links within the knowledge graph based on user intent. Consequently, this hybrid paradigm not only improves the precision of generative recommendation but also provides robust explanations by tracing the reasoning path through fused multimodal embeddings and graph topology, marking a significant shift towards more transparent and adaptable recommender systems.

2.2.2 Dynamic Context Injection and World Knowledge Reasoning

Traditional slot-based architectures rely on predefined variables to capture user intent, often failing to address the dynamic and nuanced nature of real-time interactions. In contrast, modern approaches leverage the emergent reasoning abilities of Large Language Models (LLMs) to perform dynamic context injection [10]. By formulating specialized prompts that encode recent user interactions, temporal signals, and environmental factors, these systems transform static behavioral data into rich natural language contexts. This shift allows models to move beyond frequency-based heuristics, enabling them to interpret complex, multi-

turn dialogues and adapt to evolving user preferences without requiring explicit state tracking or rigid template filling.

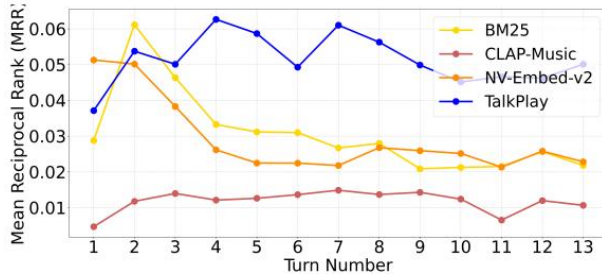


Figure 11: Chart from 'TALKPLAY: Multimodal Music Recommendation with Large Language Models' (arXiv:2502.13713)

The core advantage of this paradigm lies in the seamless integration of vast world knowledge inherent in pre-trained LLMs [1]. Unlike conventional recommenders that depend on explicit knowledge graph construction or auxiliary content enrichment, LLMs internalize fine-grained conceptual relationships regarding entities, cultural trends, and human preferences [11]. This embedded knowledge facilitates sophisticated reasoning over both structured and unstructured inputs, allowing the system to infer latent user intents and understand item semantics—such as director styles in movies or seasonal fashion trends—directly from textual descriptions. Consequently, the model can generate recommendations that are not only accurate but also contextually grounded in broader real-world understanding.

$$\mathbf{e}_s = \frac{1}{|\mathcal{T}_s|} \sum_{v \in \mathcal{T}_s} \mathbf{E}[v].$$

Figure 12: Chart from 'Fine-grained Semantics Integration for Large Language Model-based Recommendation' (arXiv:2602.22632)

To further enhance reasoning capabilities, advanced frameworks employ techniques such as Chain-of-Thought prompting and retrieval-augmented generation to refine the utilization of this world knowledge. These methods guide the model to articulate explicit reasoning trajectories, aligning its decision-making process with logical paths derived from user history and external

domain-specific information. By dynamically injecting relevant context and leveraging the model's ability to simulate plausible interactions, these systems effectively address challenges like cold-start scenarios and data sparsity. The result is a robust recommendation engine capable of performing multi-step reasoning, providing human-readable explanations, and generalizing to unseen items and domains with minimal additional training.

2.2.3 Mitigating Hallucinations via Structured Knowledge Constraints

Hallucinations in generative recommendation systems often manifest as the prediction of non-existent items or the fabrication of user preferences, primarily stemming from the decoupling between the model's parametric knowledge and the dynamic item catalog. To address this, recent methodologies impose structured knowledge constraints that restrict the generation space to valid item identifiers or verified attribute sets. By mapping latent representations directly to a predefined vocabulary of item tokens rather than open-ended text, these approaches ensure that every generated output corresponds to an existing entity within the database, effectively eliminating the possibility of inventing phantom items during the decoding phase.

Beyond token-level constraints, advanced frameworks integrate retrieval-augmented mechanisms and external memory modules to ground the reasoning process in factual interaction histories. These systems utilize structured graphs or indexed databases to retrieve relevant user behaviors and item features, which are then injected into the prompt context as hard constraints. This strategy forces the large language model to condition its predictions on retrieved evidence rather than relying solely on internal probabilistic distributions, thereby significantly reducing the incidence of factual errors and ensuring that recommended items align with the user's actual historical patterns and current intent.

Furthermore, optimization techniques such as constrained beam search and masked attention mechanisms are employed to enforce structural validity during inference. By modifying causal attention masks to prevent information leakage among candidate items and restricting the probability distribution over the output layer to a valid subset of items, these methods guarantee that the final rec-

ommendation list adheres to strict domain rules. Such structural interventions not only mitigate hallucinatory outputs but also enhance the reliability of explainable recommendations, ensuring that the generated justifications are logically consistent with the underlying data constraints and free from fabricated reasoning chains.

2.3 Agent Based Reasoning and Interactive Recommendation

2.3.1 Cognitive Architectures and Symbolic Reasoning Traces

Early cognitive architectures in recommender systems relied heavily on symbolic, slot-based state representations to model user intent. These systems populated predefined variables, such as preferred cuisine or genre, from limited value sets to drive response generation via rigid natural language templates. While effective for constrained domains, this approach suffered from severe limitations in semantic flexibility and reasoning depth. The reliance on explicit slot filling prevented the capture of nuanced, implicit user preferences, resulting in generic explanations that failed to reflect complex behavioral dynamics or provide truly personalized justification for recommendations.

The paradigm has subsequently shifted toward generative cognitive architectures that unify recommendation and reasoning within large language models. By reformulating recommendation tasks as natural language generation problems, these systems leverage pre-trained semantic knowledge to interpret unstructured interaction histories directly [12]. Advanced frameworks now employ instruction tuning and prompt engineering to guide models in synthesizing user interests from sequential data, effectively replacing static slot states with dynamic, context-aware representations. This transition enables the system to perform sophisticated inference over item metadata and user history, facilitating the generation of coherent, narrative-driven explanations that align closely with human cognitive processes.

To further enhance interpretability, recent developments integrate explicit symbolic reasoning traces, such as Chain-of-Thought prompting, into the recommendation pipeline. These techniques decompose the decision-making process into intermediate logical steps, tracing the derivation from implicit user behaviors to explicit preference formulations before arriving at a final suggestion. By

aligning the model’s internal reasoning with structured knowledge graphs or counterfactual scenarios, these architectures mitigate hallucination and ensure that generated explanations are both faithful to the underlying data and logically sound. This synthesis of neural generation and symbolic tracing represents a critical advancement in creating transparent, trustworthy, and deeply personalized recommender agents.

2.3.2 Multi Agent Collaboration and Tool Learning Frameworks

Recent frameworks have evolved from static recommendation models to dynamic multi-agent systems where autonomous entities collaborate to resolve complex user intents. In these architectures, distinct agents assume specialized roles, such as user simulators, item retrievers, and reasoning planners, interacting through structured dialogue protocols to refine preference elicitation. This collaborative paradigm enables the decomposition of multifaceted tasks into manageable sub-problems, allowing agents to leverage collective reasoning capabilities for enhanced personalization and robustness against cold-start scenarios.

Central to this evolution is the integration of tool learning mechanisms that empower agents to interact with external environments beyond their parametric knowledge. By utilizing function calling capabilities, agents can dynamically invoke databases, search engines, and domain-specific APIs to retrieve real-time information or execute transactional actions. Advanced planning strategies, often driven by few-shot prompting or reinforcement learning, allow the system to generate coherent execution sequences, tracking intermediate tool outputs to iteratively adjust recommendations and ensure alignment with evolving user constraints.

The synergy between multi-agent collaboration and tool utilization facilitates a shift towards generative, interactive recommendation experiences. Systems now employ memory modules to store long-term interaction histories, enabling agents to maintain context across multi-turn conversations while adapting to immediate feedback. This architecture not only improves the accuracy of item retrieval through collaborative filtering semantics but also enhances explainability by generating natural language rationales grounded in actual tool interactions, thereby bridging the gap between abstract reasoning and practical application in indus-

trial settings.

2.3.3 Simulated User Feedback and Self Reflective Iteration

Simulated user feedback leverages Large Language Model-based agents to emulate human decision-making processes, addressing the scarcity and noise inherent in real-world implicit feedback data. By integrating memory modules, role modeling, and contextual awareness, these agents generate coherent sequences of actions such as clicks, skips, or queries in response to recommendation slates. This approach transforms static evaluation into dynamic interaction, allowing systems to infer complex user preferences and test recommendation strategies under diverse, controlled conditions that mimic authentic cognitive and emotional responses.

Self-reflective iteration enhances this simulation by introducing feedback loops where agents critique their own generated behaviors or the system's responses to refine future interactions. Through mechanisms like generative explanation and automated critiquing, models can identify discrepancies between intended and actual user trajectories, adjusting internal states to better align with evolving preferences. This reflective capacity enables the generation of harder negative samples and more nuanced preference signals, which are critical for optimizing reinforcement learning objectives and improving the robustness of sequential recommendation models against cold-start and distribution shift challenges.

The synergy between simulated feedback and self-reflection facilitates a continuous improvement cycle for recommender systems without relying solely on costly online A/B testing. By iteratively refining prompts and incorporating synthetic conversations derived from agent interactions, systems can achieve significant gains in metrics such as Recall and NDCG. Ultimately, this paradigm shifts the focus from passive pattern matching to active, adaptive user modeling, providing a scalable framework for stress-testing algorithms and exploring long-term user intent dynamics in complex, multi-preference environments.

2.4 Preference Alignment and Controlled Generation

2.4.1 Reinforcement Learning from Human and Oracle Feedback

Reinforcement Learning from Human and Oracle Feedback (RLHF) has emerged as a pivotal strategy for aligning generative recommenders with complex user preferences and business objectives. Unlike traditional supervised fine-tuning that relies on static historical logs, RLHF frameworks leverage dynamic reward signals derived from explicit human ratings or implicit oracle feedback, such as click-through rates and conversion metrics. This paradigm shift enables models to optimize non-differentiable evaluation metrics directly, addressing the discrepancy between training losses and real-world performance indicators like Normalized Discounted Cumulative Gain (NDCG).

Recent advancements employ sophisticated algorithms, including Direct Preference Optimization (DPO) and Rank Responses to align Human Feedback (RRHF), to refine model policies without the instability often associated with traditional actor-critic methods. In these setups, reward models are trained to distinguish between superior and inferior recommendation sequences, guiding the generator to prioritize items that maximize long-term user engagement or specific revenue goals. Techniques such as list-wise preference alignment further enhance this process by considering the holistic quality of recommended lists rather than isolated item scores, effectively capturing the nuanced trade-offs between diversity, relevance, and novelty.

Furthermore, the integration of simulated user agents provides a scalable source of oracle feedback, mitigating the scarcity and cost of extensive human annotation. These cognitive agents, grounded in episodic memory and persona constraints, generate realistic interaction trajectories that facilitate robust offline policy learning and evaluation. By continuously updating user state representations based on sequential interactions, RL-driven systems can adapt to evolving intents and handle cold-start scenarios more effectively than conventional approaches. Consequently, this feedback loop not only improves recommendation accuracy but also fosters greater instruction-following capabilities, ensuring that generated explanations and suggestions remain coherent, personalized, and aligned with overarching system

goals.

2.4.2 Controllable Diversity and Serendipity Optimization

Controlling the stochastic nature of Large Language Models (LLMs) to achieve specific diversity and serendipity targets remains a formidable challenge, as standard prompt engineering and full parameter fine-tuning often fail to provide granular output regulation. While deterministic embedding approaches capture user preferences, they frequently suffer from semantic collapse and homogeneous inference mechanisms that limit the exploration of long-tail items. To address these limitations, recent methodologies integrate retrieval-augmented generation with external tool calls, enabling dynamic adaptation to shifting user intents while mitigating the risks of generating irrelevant or unsafe content through stricter boundary constraints.

Advanced optimization strategies have emerged to explicitly balance relevance with novelty, often employing reinforcement learning frameworks such as Proximal Policy Optimization to train actor networks on multi-objective reward functions. These approaches introduce specialized diversity loss terms during training to prevent the mapping of distinct items to identical semantic representations, thereby preserving the dissimilarity within recommended lists. Furthermore, consistency-calibrated fusion mechanisms combine predictions from base and tuned models, leveraging variance estimates to stabilize the generation process and ensure that increased serendipity does not come at the cost of fundamental recommendation accuracy.

Empirical evaluations indicate that multi-query generation strategies significantly enhance the comprehensiveness of user interest representation compared to single-pass inference, directly improving metrics like Diversity@K without necessitating larger candidate pools. However, simply increasing the retrieval count K without corresponding adjustments in query complexity yields diminishing returns, suggesting that search depth alone cannot induce true serendipity. Consequently, effective system design requires a synergistic combination of orthogonal transformation mechanisms for knowledge transfer and adaptive control schemes, such as cyclic KL-divergence modulation, to maintain an optimal equilibrium between exploration efficiency and convergence

stability in large-scale deployments.

2.4.3 Bias Mitigation and Fairness Aware Prompt Engineering

Bias mitigation in Large Language Model-based Conversational Recommender Systems increasingly relies on fairness-aware prompt engineering to address inherent prejudices without extensive parameter updates. Pre-trained models often internalize societal biases regarding sensitive attributes such as gender, race, and occupation, leading to skewed recommendations that marginalize minority groups. To counteract this, recent approaches focus on distilling bias into continuous soft prompts or constructing specific instruction templates that explicitly decouple user preference modeling from sensitive demographic representations. By embedding fairness constraints directly into the input context, these methods guide the model's reasoning process to prioritize equitable item scoring while maintaining recommendation accuracy.

Advanced techniques leverage counterfactual reasoning and dynamic prompt adaptation to simulate alternative exposure scenarios and neutralize position bias. For instance, prompts can be engineered to query how a user might respond under different treatment conditions, generating uplift scores that feed into inverse propensity models for debiased ranking. Furthermore, domain-specific knowledge enhancement paradigms utilize retrieval-augmented generation to inject balanced, context-aware information into the prompt, ensuring that less popular or minority viewpoints are not overshadowed by frequent patterns in the pretraining corpus. This modular approach allows for real-time intent adaptation and robust handling of cold-start scenarios where historical data is sparse or potentially biased.

Despite these advancements, challenges remain in defining universal fairness metrics and ensuring that prompt-based interventions do not inadvertently introduce new forms of bias through poorly constructed instructions [13]. Current research emphasizes the need for rigorous evaluation frameworks that assess both the veracity and personalization of generated responses across diverse demographic groups. Future directions include developing automated mechanisms for detecting fairness violations within prompt outputs and integrating multi-objective optimization strategies that balance utility with equity. As prompt engineering

evolves, it serves as a critical, interpretable layer for enforcing ethical standards, enabling low-code deployment of fair recommendation policies that enhance user trust and system transparency.

$$Recall = \frac{TP}{TP + FN}$$

Figure 13: Chart from 'Ethical AI prompt recommendations in large language models using collaborative filtering' (arXiv:2510.06924)

2.5 Efficiency Optimization and Deployment Strategies

2.5.1 Speculative Decoding and Inference Acceleration

Speculative decoding has emerged as a pivotal technique for mitigating the high inference latency inherent in large language model-based recommender systems. By decoupling the generation process into a draft phase and a verification phase, this approach utilizes a compact, lightweight draft model to propose multiple candidate tokens autoregressively. These candidates are subsequently verified in parallel by the larger target model, effectively transforming sequential token generation into a batched operation. This mechanism significantly reduces the number of forward passes required per output sequence, yielding substantial speedups while mathematically guaranteeing that the final output distribution remains identical to that of the original target model.

Recent advancements have extended basic speculative decoding through tree-based attention mechanisms and adaptive drafting strategies tailored for recommendation scenarios. Instead of linear token chains, tree-structured decoding allows the draft model to explore multiple potential reasoning paths or item sequences simultaneously, which the target model then validates using specialized attention masks. This parallel verification capability is particularly beneficial for generating diverse recommendation lists or complex explanatory narratives, where maintaining Top-K consistency is crucial. Empirical studies indicate that such optimizations can achieve acceleration factors between 2x and 2.5x without compromising the accuracy of user preference modeling or the

relevance of generated items.

Despite these gains, deploying speculative decoding in industrial recommendation pipelines introduces specific challenges regarding memory bandwidth and hardware integration. The efficiency of the verification step is often bounded by memory access speeds, especially during traffic bursts that saturate bandwidth, potentially diminishing returns from increased parallelism. Furthermore, seamless integration requires tight coupling with vector search pre-rankers and on-device neural processing units to minimize overhead. Future research directions focus on hierarchical routing schemes that combine dense retrieval with sparse Mixture-of-Experts architectures, aiming to balance tail latency reduction with the computational costs of maintaining dual-model synchronization in dynamic, real-world environments.

2.5.2 Parameter Efficient Fine Tuning and Adapter Methods

Parameter-Efficient Fine-Tuning (PEFT) has emerged as a critical paradigm for adapting Large Language Models to recommendation systems, addressing the prohibitive computational costs associated with full-model weight updates [4]. Given the latency-sensitive nature of recommender applications and the massive scale of underlying language backbones, traditional fine-tuning often proves infeasible for resource-constrained environments. PEFT methodologies circumvent this by freezing the majority of pre-trained parameters and optimizing only a small, task-specific subset, thereby significantly reducing GPU memory requirements and enabling deployment on single consumer-grade graphics processors while maintaining competitive performance.

Adapter-based approaches, particularly Low-Rank Adaptation (LoRA), represent a dominant strategy within this domain by injecting trainable low-rank decomposition matrices into the transformer layers. This architecture allows for efficient alignment of general-purpose language models with specific recommendation objectives, such as item ranking or sequential prediction, without altering the original model weights. By restricting gradient updates to these lightweight adapter modules, researchers have achieved substantial reductions in training time and storage overhead, facilitating rapid iteration and the potential for maintaining distinct adapters for diverse user segments or

domains within a shared backbone infrastructure.

Complementary to adapter methods, prompt tuning techniques optimize learnable soft prompts or continuous embeddings rather than model parameters, leveraging frozen LLMs to generate high-quality recommendations [14]. These strategies are often combined with instruction tuning frameworks to enhance the model’s ability to follow complex recommendation directives and interpret user intent accurately. While PEFT methods drastically lower the barrier for entry, ongoing research focuses on further improving adaptation efficiency against rapidly expanding datasets and exploring hybrid architectures that balance the trade-offs between parameter count, inference latency, and recommendation accuracy in large-scale industrial deployments.

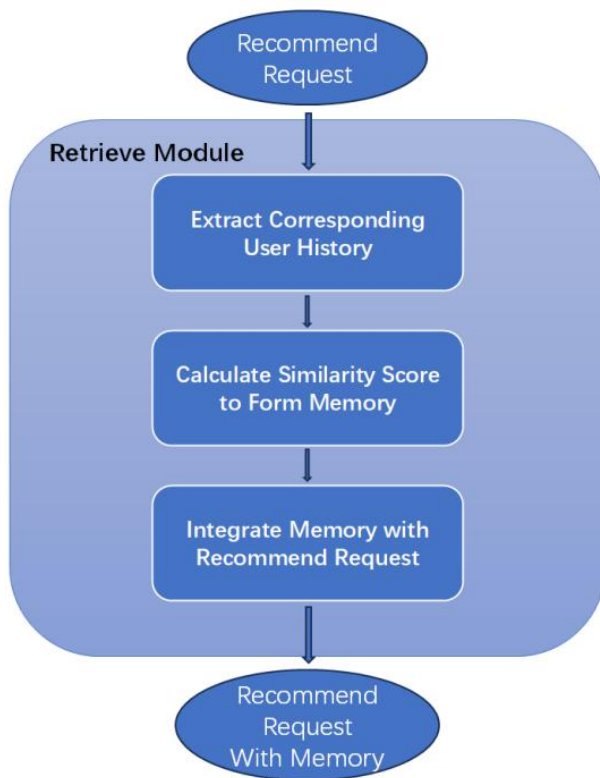


Figure 14: Chart from 'Memory Assisted LLM for Personalized Recommendation System' (arXiv:2505.03824)

2.5.3 Federated Learning and Privacy Preserving Architectures

Federated Learning (FL) has emerged as a pivotal architecture for deploying Large Language Model (LLM)-based recommender systems while strictly adhering to privacy constraints. By decentralizing model training, FL ensures that sensitive user

interaction data remains on local devices, transmitting only model updates or gradients to a central server. This paradigm is particularly critical for LLMs, which risk memorizing private behavioral sequences during centralized fine-tuning. Recent adaptations integrate FL with collaborative filtering mechanisms, enabling the aggregation of global semantic knowledge without exposing individual user histories, thus mitigating the risk of data leakage inherent in traditional cloud-centric training pipelines.

To further fortify privacy guarantees, contemporary architectures frequently couple FL with Differential Privacy (DP) mechanisms. This combination introduces calibrated noise to the local updates before aggregation, providing rigorous mathematical assurances against inference attacks and membership reconstruction. While federated matrix factorization and prompt learning strategies have demonstrated promising privacy-utility trade-offs, they introduce significant technical challenges, including increased communication overhead, model drift across heterogeneous clients, and the complexity of managing non-IID data distributions. Specialized masking strategies and secure aggregation protocols are often employed to prevent information leakage during the multi-round synchronization processes required for converging large-scale language models.

Despite these advancements, the integration of FL into LLM-driven recommendation entails balancing computational efficiency with robust security. Emerging approaches explore federated prompt learning and synthetic user simulations to reduce the communication burden while maintaining personalization accuracy. These methods aim to achieve performance comparable to centralized training by leveraging local inductive capabilities without compromising user-level protections. However, the deployment of such systems requires specialized infrastructure to handle the latency and bandwidth demands of transmitting high-dimensional LLM parameters, marking a critical area for future optimization in trustworthy, privacy-preserving recommendation ecosystems.

3 Sequential User Behavior Modeling and Retrieval

3.1 Lifelong Behavior Modeling and Interest Evolution

3.1.1 Hierarchical Summarization of Long Term Sequences

Handling long-term user interaction sequences presents significant challenges for Large Language Models due to context window limitations and the noise introduced by heterogeneous behaviors. Traditional sequential recommendation approaches often rely on chronological ordering, which retains irrelevant historical interactions that obscure the user’s current intent [15]. To mitigate this, hierarchical summarization techniques have emerged as a critical strategy to compress extensive behavioral histories into concise, semantically dense representations. By aggregating homogeneous behaviors similar to a target item while filtering out unrelated activities, these methods effectively reduce the heterogeneity score of the input sequence, thereby enhancing the model’s comprehension capabilities.

The core mechanism involves transforming raw interaction logs into structured summaries through multi-level abstraction. Instead of processing every individual item token, advanced frameworks employ semantic retrieval to identify top-relevant behaviors, constructing sequences that prioritize relevance over recency. This process often utilizes discretized indexing or clustering algorithms to group items based on shared semantic features, creating a hierarchical structure where lower levels capture fine-grained details and higher levels represent aggregated user preferences. Such an approach ensures that the resulting embeddings capture both individual item characteristics and broader sequential patterns, allowing the model to focus on high-order dependencies without being overwhelmed by voluminous data.

Empirical evidence suggests that reducing sequence heterogeneity directly correlates with improved recommendation performance, particularly as the sequence length increases [16]. By homogenizing the input through hierarchical summarization, Large Language Models can more effectively extract useful signals from long-term history, alleviating the incomprehension problems associated with noisy, diverse data streams. This paradigm shift from processing raw chronological lists to analyzing summarized, relevance-based hierarchies

enables more accurate prediction of future interactions. Consequently, integrating these summarization techniques serves as a foundational step for subsequent prompt-level and parameter-level enhancements, ensuring robust scalability in personalized recommendation tasks.

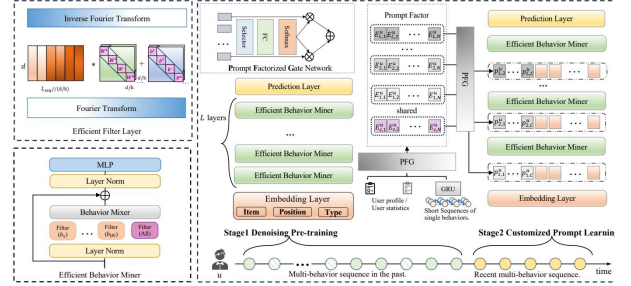


Figure 15: Chart from 'Denoising Pre-Training and Customized Prompt Learning for Efficient Multi-Behavior Sequential Recommendation' (arXiv:2408.11372)

3.1.2 Disentangling Short Term Intent from Long Term Persona

Disentangling short-term intent from long-term persona remains a critical challenge in LLM-based recommendation, as models often struggle to comprehend extensive behavioral sequences even within their context windows [17]. Traditional approaches frequently rely on heuristic assumptions or simple inner-product matching, which fail to capture the nuanced, multi-faceted nature of user interests. Consequently, LLMs may over-emphasize frequent interactions while underrepresenting diverse contextual signals, leading to homogeneous representations that obscure individualistic preferences and hinder the extraction of essential knowledge from long history.

To address this, recent methodologies employ hierarchical modeling and sequence segmentation to decouple item-level semantics from user-level evolution. By compressing complex textual descriptions into concise embeddings via an Item LLM before feeding them into a User LLM, these frameworks reduce sequence length while preserving semantic richness. Furthermore, advanced techniques segment full historical behaviors into clustered sub-sequences, applying offline summarization or reflection to generate fine-grained personas. This process allows the system to isolate transient consumption habits from stable underlying traits, effectively mitigating the risk of hallucinations that increase as interaction sequences

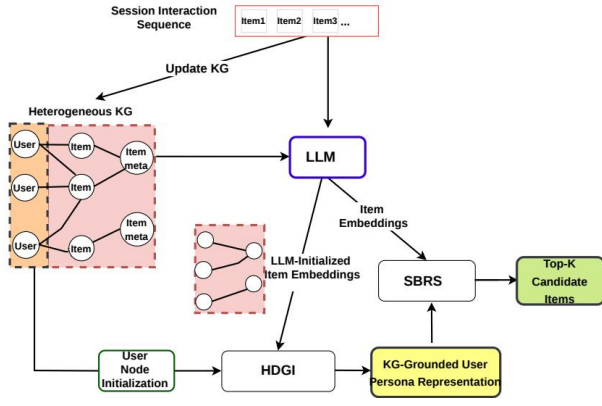


Figure 16: Chart from 'Leveraging LLMs and Heterogeneous Knowledge Graphs for Persona-Driven Session-Based Recommendation' (arXiv:2604.06928)

shorten or become sparse.

The integration of exogenous behavioral and semantic signals further enhances this disentanglement by facilitating dual information streams that mutually reinforce comprehension. Instead of treating user history as a monolithic block, these approaches reconstruct prior preference distributions by parsing latent categories and inter-item relationships. Through contrastive objectives and self-distillation mechanisms, models learn to align projected intent embeddings with target-domain structures, filtering out noise from irrelevant source information. This synergy ensures that the system captures both immediate engagement drivers and enduring user identities, ultimately improving generalization ability across diverse interaction scenarios.

3.1.3 Temporal Dynamics and Cross Domain Interest Transfer

Modeling temporal dynamics in sequential recommendation requires capturing the evolution of user interests across varying time scales [18]. Traditional approaches often rely on recurrent or self-attention mechanisms to encode chronological sequences, yet they frequently struggle with the incomprehension of lifelong behavioral patterns due to noise and sparsity. To address this, recent advancements incorporate retrieval-enhanced strategies that substitute raw historical sequences with semantically relevant behaviors, preserving chronological order while reducing heterogeneity. This augmentation allows models to better isolate dominant signals from latent interests, mitigating the overemphasis on recent interactions that typically plagues purely collaborative filtering meth-

ods.

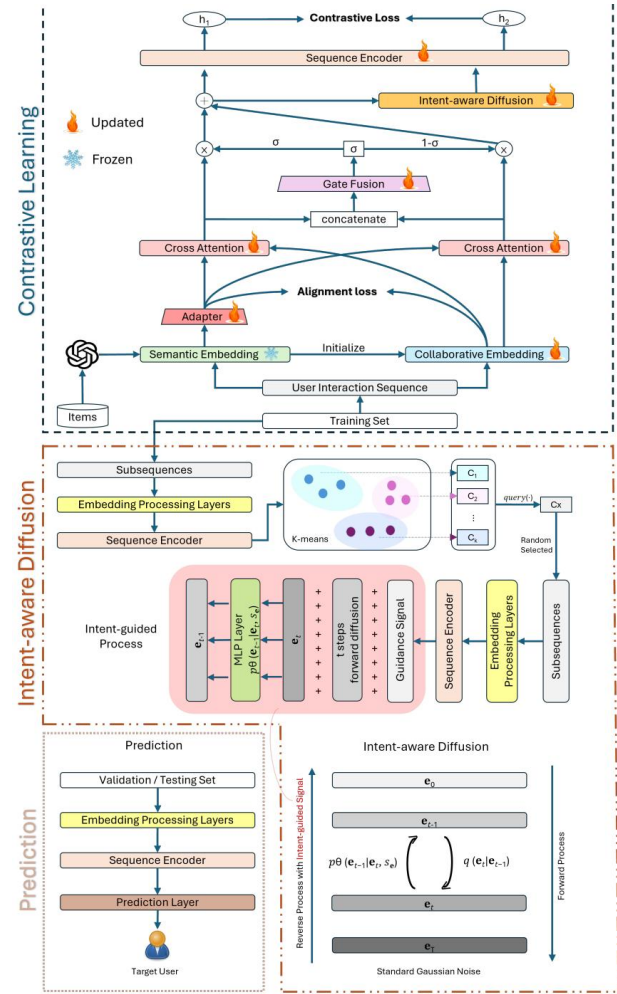


Figure 17: Chart from 'LLMDiRec: LLM-Enhanced Intent Diffusion for Sequential Recommendation' (arXiv:2601.03259)

Cross-domain interest transfer further complicates temporal modeling by necessitating the alignment of user preferences across heterogeneous item spaces. Effective transfer relies on identifying shared semantic structures rather than superficial interaction overlaps, often leveraging auxiliary information such as knowledge graphs or item attributes. However, static auxiliary sources may fail to capture dynamic multi-interest shifts in real-world scenarios. Consequently, unified frameworks now employ dual-view modeling and contrastive learning to disentangle domain-specific noise from invariant user intent, enabling the propagation of interest patterns from source to target domains without relying on extensive external APIs.

The integration of large language models offers a promising avenue for resolving these challenges

by encoding descriptive texts into unified representations. By treating user behavior sequences as textual narratives, these models can perform offline bulk inference to predict novel interest transitions, significantly reducing online computational overhead. This paradigm shift moves expensive inference stages offline, utilizing pre-computed interest clusters for efficient online serving. Ultimately, combining retrieval-augmented sequence construction with cross-domain semantic alignment fosters a robust mechanism for understanding complex temporal dynamics and facilitating scalable, personalized recommendations across diverse domains.

3.2 Semantic Enhancement and Multimodal Sequence Fusion

3.2.1 Integrating Textual Semantics with Behavioral IDs

Integrating textual semantics with behavioral IDs addresses the fundamental disconnect between the discrete, collaborative nature of traditional recommendation identifiers and the continuous, semantic reasoning capabilities of Large Language Models. While IDs effectively capture user-item interaction patterns, they lack intrinsic meaning for LLMs, which rely on tokenized text to access world knowledge [19]. Consequently, naive replacement of IDs with item titles often fails to preserve the nuanced collaborative signals essential for personalized ranking, creating a semantic gap that hinders the model’s ability to correlate historical behaviors with candidate items effectively.

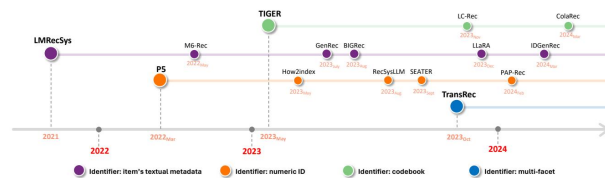


Figure 18: Chart from ‘A Survey of Generative Search and Recommendation in the Era of Large Language Models’ (arXiv:2404.16924)

To bridge this disparity, recent methodologies employ specialized injection mechanisms that map discrete IDs into the LLM’s embedding space without discarding their collaborative integrity. Instead of relying solely on standard lookup tables or raw text descriptions, these approaches utilize learnable ID injection components or vector quantization techniques to generate semantic IDs. These identifiers are designed such that semanti-

cally similar items share prefix structures, allowing the model to implicitly learn item correlations and latent categories. By concatenating these projected ID embeddings with textual prompts at the input layer, the architecture ensures that both collaborative filtering signals and rich semantic attributes are processed simultaneously within the transformer layers.

Furthermore, advanced frameworks enhance this integration by aligning user-specific behavioral sequences with personalized semantic indices through customized pre-training tasks. This process involves reconstructing prior preference distributions from interaction sequences, forcing the LLM to parse the latent relationships between different IDs while leveraging external knowledge for reasoning. By maintaining the structural efficiency of ID-based modeling while infusing it with the explanatory power of natural language, these hybrid architectures achieve a robust synthesis of collaborative and semantic information. This dual-modality approach not only mitigates data sparsity in long-tail scenarios but also provides a scalable foundation for generating precise, context-aware recommendations.

3.2.2 Multimodal Alignment in Sequential Contexts

Multimodal alignment in sequential contexts addresses the critical challenge of harmonizing heterogeneous data streams, such as textual semantics and behavioral IDs, within dynamic user interaction histories. Traditional sequential models often struggle to integrate dense collaborative signals with sparse semantic descriptions due to modality gaps and varying temporal granularities. Recent advancements leverage transformer-based architectures to project disparate modalities into a unified latent space, ensuring that semantic clusters derived from item descriptions align coherently with collaborative interests inferred from interaction sequences.

To achieve effective alignment, contemporary frameworks employ sophisticated attention mechanisms and projection layers that dynamically allocate semantic features to specific collaborative interests. These methods often utilize capsule networks or contrastive learning objectives to capture multi-interest representations without suffering from over-coarse or over-fine clustering. By reconstructing exogenous signals through non-invasive multiscale tasks, models can decompress

knowledge-rich tokens and integrate them with historical behavior patterns, thereby enhancing the comprehension of complex, long-term user preferences that vanilla adaptation techniques frequently miss.

Furthermore, the alignment process is optimized through specialized reconstruction tasks that facilitate deep integration beyond simple next-item prediction. These strategies include global contrast decomposition and comprehensive interaction modeling, which allow the backbone language model to assimilate high-compression semantic signals efficiently. By freezing certain embedding layers while fine-tuning alignment modules, these approaches mitigate optimization difficulties arising from unbalanced training stages, ultimately enabling robust sequential reasoning that scales effectively with increasing sequence lengths and multimodal complexity.

3.2.3 Denoising Sparse Interactions via Semantic Augmentation

Data sparsity remains a critical bottleneck in sequential recommendation, where limited user-item interactions hinder the accurate modeling of preferences. To mitigate this, recent paradigms leverage Large Language Models (LLMs) to inject rich world knowledge into sparse collaborative signals through semantic augmentation. Unlike traditional data augmentation techniques that rely on heuristic perturbations or masking strategies, LLM-driven approaches generate contextually precise semantic representations that preserve item independence while enriching feature spaces. This process effectively supplements scarce interaction data by inferring latent categories and item relationships, thereby constructing a more robust foundation for preference learning without introducing the erroneous semantics often associated with direct LLM hallucinations.

The core mechanism involves transforming raw interaction sequences into dense semantic embeddings via pre-trained language backbones, which are then fused with traditional ID-based representations. Advanced frameworks employ multi-level attention mechanisms to dynamically aggregate these augmented signals, allowing specific semantic cues to dominate cluster representations and sharpen multi-interest encoding. By utilizing techniques such as residual-quantized variational autoencoders, models can hash user information into discrete personalized semantic IDs within a uni-

fied space. This fusion not only captures high-order feature interactions explicitly but also facilitates efficient similarity-based retrieval, ensuring that the enhanced representations remain compatible with limited textual information while maximizing the utility of exogenous knowledge.

Furthermore, semantic augmentation serves as an implicit denoising filter by distinguishing between transient noise and genuine long-term preferences. Through contrastive learning tasks and global contrast decomposition, these methods minimize the domain gap between natural language semantics and collaborative filtering signals. The integration of holistic attention enhancement ensures that retrieved semantic information acts as contextual guidance rather than direct supervision, effectively circumventing potential noise from over-coarse LLM analysis. Consequently, this approach significantly improves model robustness on extremely sparse data groups, demonstrating superior performance in capturing complex item transitions and stabilizing recommendations where traditional collaborative signals are insufficient or misleading.

3.3 Retrieval Enhanced Sequential Recommendation

3.3.1 Behavior Aware Retrieval and Context Homogenization

Traditional sequential recommendation models often rely on truncating user history to the most recent K interactions, a strategy that frequently discards diverse, latent interests embedded in older behaviors while retaining noisy or irrelevant clicks [20]. This approach exacerbates sequence heterogeneity, making it difficult for Large Language Models (LLMs) to distill coherent preference signals from disjointed behavioral patterns. Consequently, purely collaborative modeling based on recency often yields suboptimal profiling, as the input context fails to reflect the semantic consistency required for effective reasoning and knowledge extraction by generative backbones.

To mitigate these limitations, Behavior Aware Retrieval and Context Homogenization mechanisms replace naive truncation with Semantic User Behavior Retrieval (SUBR). Instead of selecting items based solely on temporal proximity, SUBR identifies and constructs input sequences comprising the top- K behaviors that are semantically most relevant to the target item. This retrieval

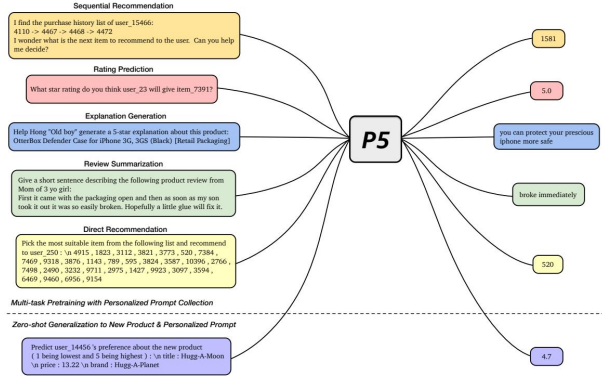


Figure 19: Chart from 'Recommendation as Language Processing (RLP): A Unified Pretrain, Personalized Prompt & Predict Paradigm (P5)' (arXiv:2203.13366)

process effectively filters out random noise and aligns the historical context with the current prediction task, significantly lowering the heterogeneity score of the input sequence. By curating a behavior sequence that is semantically homogeneous, the method ensures that the LLM receives high-quality, focused data samples that facilitate the extraction of meaningful user intent [21].

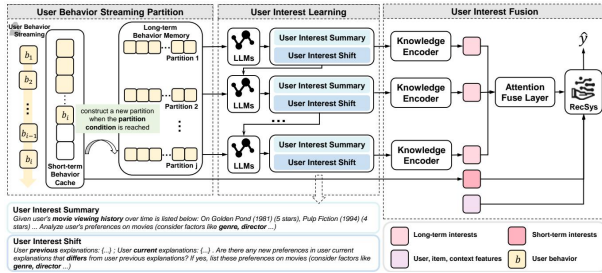


Figure 20: Chart from 'LIBER: Lifelong User Behavior Modeling Based on Large Language Models' (arXiv:2411.14713)

The integration of behavior-aware retrieval fundamentally transforms the input landscape for LLM-driven recommenders, shifting from raw, heterogeneous logs to refined, task-specific contexts. This homogenization not only addresses the context length constraints inherent in processing long textual behaviors but also enhances the model's ability to generalize across different interaction scenarios. Empirical evidence suggests that such semantically constructed sequences enable LLMs to achieve superior performance compared to baseline methods relying on recent-only inputs, demonstrating that improving data quality through targeted retrieval is critical for unlocking the full potential of language models in sequential recommendation tasks [22].

$$\mathcal{L}_{SRS}^A = - \sum_{u \in \mathcal{U}} \sum_{j=1}^{n_A} \log \sigma(P(v_{j+1} = v^+) - P(v_{j+1} = v^-))$$

Figure 21: Chart from 'Bridge the Domains: Large Language Models Enhanced Cross-domain Sequential Recommendation' (arXiv:2504.18383)

3.3.2 Collaborative Signal Infusion into Prompt Contexts

Collaborative signal infusion into prompt contexts addresses the inherent disconnect between the semantic reasoning capabilities of pre-trained large language models and the latent relational patterns found in user-item interactions. Traditional prompting strategies often rely solely on textual descriptions or explicit historical sequences, neglecting the rich collaborative filtering signals that define community-based preferences. To bridge this gap, recent methodologies inject collaborative knowledge directly into the input context, transforming discrete interaction IDs or aggregated behavioral vectors into natural language templates. This process aligns the LLM's semantic space with the recommendation domain, enabling the model to interpret non-textual collaborative cues as contextual instructions rather than opaque numerical inputs.

Advanced techniques employ soft prompt augmentation and decompression guidance projectors to seamlessly integrate these external signals without disrupting the model's pre-trained knowledge. Instead of rigidly concatenating raw ID sequences, which can cause tokenization inefficiencies and semantic misalignment, these approaches map collaborative representations into continuous latent states that guide the attention mechanism. By utilizing multi-level attention structures, the model first captures core interaction patterns and then refines them with holistic contextual information retrieved from similar users or items. This parameter-efficient strategy allows the LLM to leverage exogenous behavioral data for next-item prediction while mitigating the risk of hallucinations often associated with direct supervision from noisy collaborative graphs.

The efficacy of infusing collaborative signals is particularly pronounced in cold-start and sparse data scenarios, where explicit textual features are insufficient to characterize user intent. By conditioning the generation process on distilled collaborative profiles, such as normalized interaction

frequency vectors or cluster-based interest summaries, the model can infer latent cross-category interests that purely sequential modeling might miss. This paradigm shifts the role of the LLM from a passive text processor to an active reasoner that synthesizes semantic content with structural behavioral data. Consequently, the resulting prompts facilitate more robust generalization across diverse user groups, ensuring that recommendations reflect both the semantic relevance of items and the underlying collaborative dynamics of the user base.

3.3.3 Efficient Caching and Offline Profile Generation

To mitigate the prohibitive latency associated with real-time LLM inference in recommendation systems, recent frameworks advocate for an offline profile generation paradigm. This approach shifts the computational burden from online serving to a pre-processing stage, where extensive user behavioral sequences are analyzed to synthesize compact, semantically rich user profiles. By leveraging techniques such as max covering problems to ensure synthesized users exhibit compact yet representative interests, systems can construct high-fidelity core-sets that preserve the full spectrum of user preferences without requiring on-the-fly generation during inference.

Once generated, these personalized profiles are stored in efficient caching mechanisms, enabling downstream recommendation agents to retrieve pre-computed representations rather than invoking heavy generative processes for each request. This retrieval-based architecture significantly reduces online inference overhead and eliminates the latency bottlenecks inherent in processing long behavioral contexts dynamically. Furthermore, by decoupling profile modeling from immediate decision-making, the system ensures that complex reasoning tasks regarding user preferences are resolved offline, allowing the online module to focus solely on rapid matching and ranking based on cached embeddings.

The efficacy of this strategy is further enhanced by integrating parameter-efficient fine-tuning methods, such as Low-Rank Adaptation, to optimize the profile generation models without updating the entire backbone. This combination of offline synthesis and cached retrieval not only addresses the challenge of lifelong sequence incomprehension but also ensures scalability in

industrial deployments where user behavior data vastly exceeds item catalogs. Consequently, this dual-phase architecture achieves a superior balance between recommendation accuracy and computational efficiency, making it viable for large-scale applications with strict latency constraints.

3.4 Robustness and Adversarial Defense in Sequences

3.4.1 Detecting and Mitigating Poisoning Attacks

Poisoning attacks in Large Language Model-based recommendation systems (LLM4Rec) typically involve injecting malicious interaction sequences or manipulating prompt structures to skew model predictions toward target items. Unlike traditional collaborative filtering vulnerabilities, these attacks exploit the semantic understanding and generative capabilities of LLMs, often embedding adversarial patterns within natural language descriptions or ID token sequences. The severity is exacerbated by the open-ended nature of prompt inputs, where subtle perturbations in user history or item descriptions can propagate through attention mechanisms, causing significant degradation in recommendation fairness and accuracy while remaining stealthy to standard validation protocols.

To detect such threats, recent methodologies employ statistical anomaly detection on embedding spaces and monitor gradient updates during parameter-efficient fine-tuning. Techniques include analyzing the divergence between expected and actual attention weights when processing suspicious prompts, as well as utilizing clustering algorithms to identify outlier interaction patterns that deviate from established user behavioral norms. Some approaches integrate dedicated detector modules that evaluate the consistency of generated rationales against historical user preferences, flagging instances where the model's reasoning exhibits abrupt shifts indicative of poisoned training signals or adversarial inference attempts.

Mitigation strategies primarily focus on robust training objectives and input sanitization mechanisms designed to neutralize adversarial influences without compromising recommendation utility. Defenses often involve adversarial training loops where the model is exposed to synthesized poisoning scenarios to learn invariant representations, alongside dynamic prompt filtering that

scrubs potentially malicious tokens before they enter the encoder. Furthermore, differential privacy techniques and gradient clipping are increasingly adopted during the fine-tuning phase to limit the impact of any single poisoned sample on the global model parameters, ensuring that the system maintains stability and reliability even under active attack conditions.

3.4.2 Handling Noisy Interactions and False Feedback

Handling noisy interactions and false feedback remains a critical challenge in sequential recommendation, as user behavior logs often contain accidental clicks, bot-generated traffic, or transient interests that distort preference modeling. Traditional approaches frequently rely on attention mechanisms or memory networks to implicitly down-weight irrelevant items within interaction sequences. However, these methods often assume that noise follows a specific distribution or that clean signals can be easily separated through local context aggregation, which may not hold in highly sparse or adversarial environments where noise patterns are complex and non-stationary.

Recent advancements have shifted towards explicit noise-decoupling frameworks and robust learning paradigms to mitigate these issues. Techniques such as curriculum disentangled recommendation and efficient noise-decoupling modules attempt to separate multi-behavior signals, treating implicit feedback noise as a distinct component rather than mere outliers. Furthermore, the integration of large language models has introduced novel pseudo-interaction generation strategies, where cyclic batch supervision and alignment losses are employed to filter semantic noise and correct feature deviations inherent in augmented data, thereby enhancing the model’s resilience against corrupted training signals.

Despite these progressions, significant limitations persist regarding the robustness of current systems under varying noise intensities and adversarial attacks. Empirical analyses indicate that simple denoising methods can exhibit unstable performance, occasionally exacerbating attack success rates when faced with targeted promotion strategies. Moreover, many frameworks rely on strong assumptions about data independence or the availability of sufficient overlapping interactions for cross-domain supervision. When these assumptions are violated, particularly in out-of-

distribution scenarios with sparse historical data, model performance degrades, highlighting the need for more adaptive mechanisms capable of dynamically refining user profiles without overfitting to artificial or malicious feedback patterns.

3.4.3 Robust Representation Learning under Distribution Shifts

Robust representation learning under distribution shifts addresses the critical challenge of maintaining model performance when training and testing data originate from divergent distributions. Traditional approaches often rely on domain adaptation techniques that learn invariant features across sources, yet these methods frequently struggle with severe covariate shifts inherent in dynamic environments like sequential recommendation. Recent advancements have shifted towards leveraging contrastive learning frameworks to enforce consistency between augmented views of the same instance, thereby capturing correlations among items and sub-sequences that remain stable despite environmental perturbations.

To further enhance resilience, modern architectures integrate diffusion models and adversarial training to simulate and mitigate distributional noise. By reconstructing features from injected noise or generating synthetic samples that bridge domain gaps, these models learn to fit complex data distributions more effectively than standard deterministic encoders. Specifically, techniques involving random noise injection with bounded norms and domain-specific item replacement allow the model to generalize better by explicitly optimizing against worst-case perturbations, ensuring that the learned embeddings retain semantic integrity even when faced with unseen data patterns.

Furthermore, the incorporation of large language models facilitates a unified text-to-text paradigm that inherently supports transfer learning across disparate domains [23]. By mapping heterogeneous inputs into a shared semantic space, these frameworks reduce the reliance on domain-specific priors and enable zero-shot or few-shot adaptation capabilities. The synergy between low-rank adaptation mechanisms and robust objective functions allows for efficient fine-tuning, ensuring that the backbone representations remain discriminative while adapting to new distributional characteristics without catastrophic forgetting or significant computational overhead.

$$\mathcal{L} = - \sum_{i=1}^N y_i \log(\hat{y}_i)$$

Figure 22: Chart from 'E4SRec: An Elegant Effective Efficient Extensible Solution of Large Language Models for Sequential Recommendation' (arXiv:2312.02443)

3.5 Personalized Adaptation and Federated Sequential Learning

3.5.1 Dynamic LoRA Aggregation for User Specific Patterns

Dynamic LoRA aggregation addresses the limitations of static parameter adaptation by constructing personalized low-rank matrices tailored to individual user behavior patterns. Unlike vanilla LoRA, which applies a fixed update across all inputs, this approach generates multiple sets of LoRA parameters and utilizes a lightweight recommendation model to compute dynamic scaling weights. These weights aggregate distinct parameter sets based on the specific semantic context of the user's interaction sequence, allowing the large language model to flexibly adjust its representational capacity without modifying the frozen backbone architecture.

The mechanism fundamentally reinterprets the low-rank update as a weighted sum of interacting matrix pairs rather than a single static product. By decomposing the adaptation into several components, the system can capture heterogeneous behavioral signals, such as shifting interests or diverse item categories, within long user sequences. This fine-grained control enables the model to homogenize relevant behaviors while filtering out noise, effectively enhancing the comprehension of sparse or complex interaction histories through an adaptive alignment of semantic features and collaborative signals.

Consequently, this dynamic strategy significantly improves performance in scenarios characterized by data sparsity and long-tail distributions. By synthesizing user-specific adapters on the fly, the framework bridges the gap between global pre-trained knowledge and local user preferences, ensuring that the model remains sensitive to individ-

ual nuances. This approach not only boosts recommendation accuracy but also maintains computational efficiency, as it leverages the inherent parameter efficiency of low-rank decomposition while introducing a negligible overhead for the dynamic weight generation process.

3.5.2 Privacy Preserving Behavioral Augmentation

Privacy-preserving behavioral augmentation leverages Large Language Models to synthesize realistic user interaction sequences without exposing raw historical data. By employing differentially private mechanisms, these frameworks generate synthetic queries and behavioral tokens that statistically mirror original user preferences while adhering to strict privacy budgets. This approach effectively decouples the utility of behavioral data from the risk of re-identification, allowing models to learn from augmented datasets that maintain the structural integrity of long-term user interests without compromising individual anonymity.

A critical technical challenge in this domain involves mitigating LLM hallucinations that may distort semantic consistency or introduce bias into the generated sequences [24]. Advanced methodologies address this by integrating retrieval-augmented generation strategies, where semantically relevant historical behaviors are retrieved to condition the synthesis process, thereby denoising the input and ensuring fidelity to ground-truth preferences. Furthermore, parameter-efficient fine-tuning techniques, such as Low-Rank Adaptation, are utilized to align the generative capabilities of pre-trained models with specific collaborative filtering signals, ensuring that the augmented data accurately reflects complex, multi-faceted user intents rather than generic patterns.

$$\mathcal{L} = - \frac{1}{|\mathcal{B}|} \sum_j [y_j \log(\hat{y}_j) + (1 - y_j) \log(1 - \hat{y}_j)],$$

$$\hat{y}_j = \sigma(\mathbf{o} \cdot \mathbf{i}_{j,all})$$

Figure 23: Chart from 'Enhancing Sequential Recommendation with World Knowledge from Large Language Models' (arXiv:2511.20177)

The integration of these synthetic behaviors significantly alleviates data sparsity issues, particularly for cold-start users or those with fragmented interaction histories. By expanding the training corpus with high-quality, privacy-compliant

pseudo-interactions, recommendation systems can achieve robust generalization across diverse user groups without relying on external APIs or sensitive feature exposure. This paradigm not only enhances model performance on ranking metrics but also establishes a scalable foundation for trustworthy recommendation ecosystems, balancing the competing demands of data utility, computational efficiency, and rigorous privacy preservation in real-world deployments.

3.5.3 Orthogonal Feature Learning in Distributed Settings

Orthogonal feature learning in distributed settings addresses the critical challenge of integrating diverse semantic and collaborative signals without inducing catastrophic forgetting or excessive communication overhead. By decoupling behavioral and semantic feature spaces, recent frameworks avoid the compression losses inherent in encoder-side fusion, instead employing discrete indexing schemes to splice exogenous signals. This orthogonality ensures that domain-specific user behaviors are learned independently from global knowledge, allowing models to maintain robustness against distribution shifts across heterogeneous nodes while preserving the integrity of pre-trained representations.

To achieve computational efficiency in these distributed environments, low-rank adaptation techniques have become paramount for fine-tuning large language models [25]. By parameterizing weight updates as the product of two small matrices with a rank significantly lower than the original dimensions, training overhead can be reduced by up to seventy percent. This approach freezes the massive pre-trained backbone while only optimizing lightweight linear projection components, facilitating affordable updates across distributed clusters. Such strategies enable the seamless injection of collaborative information into the model architecture without modifying the underlying structure, making them ideal for resource-constrained edge devices.

$$\mathcal{L}_D = \frac{1}{|\mathcal{T}|} \sum_{(u,i) \in \mathcal{T}} \frac{1}{|E_{u,i}|} \sum_{t=1}^{|E_{u,i}|} -\log c_{|F_{u,i}|+t}^{e_t}$$

Figure 24: Chart from 'A Survey on Large Language Models for Personalized and Explainable Recommendations' (arXiv:2311.12338)

Furthermore, retrieval-augmented mechanisms complement orthogonal learning by dynamically supplementing sparse interaction data with nearest-neighbor information aggregated via attention mechanisms. In distributed scenarios, this allows local nodes to leverage global embeddings through efficient cosine similarity searches, enhancing feature representation without transmitting raw data. The combination of orthogonal decomposition, low-rank updates, and retrieval augmentation creates a scalable paradigm where local personalization and global knowledge coexist. This synergy not only mitigates the latency issues associated with full model synchronization but also ensures that the learned features remain distinct yet complementary, ultimately improving recommendation accuracy in web-scale systems.

4 Foundation Model Pre-training and Decoding Strategies

4.1 Advanced Pre-training Objectives for Behavior Sequences

4.1.1 Beyond Next Token Prediction and Multi-Token Modeling

Traditional autoregressive next-token prediction, while foundational, often struggles with the sequential nature of recommendation tasks where items map to multiple tokens. This standard paradigm treats each token generation step independently, potentially ignoring the holistic structure of an item and leading to exposure bias during inference. Furthermore, strictly adhering to single-token forecasting fails to capture the diminishing uncertainty as prefix tokens are established, resulting in inefficient decoding processes that do not fully leverage the deterministic relationships within an item's token sequence.

To address these limitations, multi-token modeling strategies have emerged to predict entire item sequences or multiple future tokens simultaneously [26]. By establishing a bijective mapping between items and their corresponding k-token sequences, recent theoretical work demonstrates that k-token autoregressive prediction is mathematically equivalent to full-item-vocabulary maximum likelihood estimation. This equivalence validates approaches that move beyond incremental generation, allowing models to forecast complete item representations in parallel or cascaded manners, thereby aligning the training objective more closely with the ultimate goal of item-level recom-

mentation rather than mere token completion.

$$\text{PPL}(y) = \exp \left(-\frac{1}{T} \sum_{t=1}^T \log p(y_t | y_{<t}) \right).$$

Figure 25: Chart from 'Generative Conversational Recommender System' (arXiv:2605.21987)

Advanced techniques further refine this approach through speculative decoding and token-level weighting mechanisms. Speculative decoding accelerates inference by drafting multiple future tokens via a smaller model and verifying them in parallel, significantly reducing large language model calls while maintaining output distribution fidelity [27]. Additionally, adaptive weighting schemes adjust the emphasis on different tokens within a sequence, reflecting the varying information density from early to late tokens. These innovations collectively enhance behavior sequence modeling, mitigating long-tail distribution issues and improving both the efficiency and accuracy of user behavior prediction in complex, real-world scenarios.

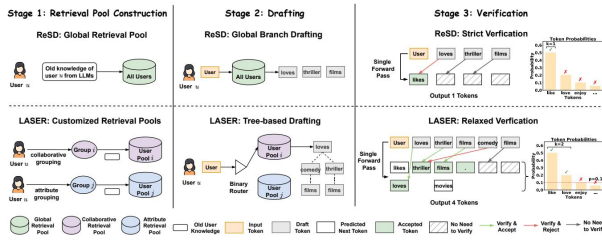


Figure 26: Chart from 'Efficiency Unleashed: Inference Acceleration for LLM-based Recommender Systems with Speculative Decoding' (arXiv:2408.05676)

4.1.2 Causal and Contrastive Pre-training Tasks

Causal pre-training tasks in sequential recommendation aim to disentangle genuine user intent from spurious correlations induced by popularity or exposure bias. By constructing causal graphs that model the interaction between historical behavior sequences, latent knowledge, and prediction outcomes, these methods employ do-calculus interventions to isolate the direct effect of user preferences. This approach often involves reweighting training samples via inverse propensity scoring or utilizing counterfactual reasoning to ensure the model learns robust representations that generalize beyond observed data distributions, effectively

mitigating confounding variables inherent in implicit feedback logs.

Complementing causal mechanisms, contrastive pre-training tasks leverage self-supervised learning to refine item and user embeddings by maximizing agreement between positive pairs while pushing negative samples apart in the latent space. Inspired by Contrastive Predictive Coding, these objectives typically involve augmenting user behavior sequences through masking, cropping, or reordering to create diverse views of the same underlying intent. The model is then trained to distinguish the true subsequent item or masked token from a set of distractors, forcing the encoder to capture high-level semantic structures and temporal dependencies rather than memorizing superficial co-occurrence patterns.

The integration of causal and contrastive objectives creates a synergistic effect where causal interventions provide unbiased structural constraints, and contrastive learning enhances the discriminative power of the representations. In practice, this is often realized through multi-task frameworks where a primary next-item prediction loss is regularized by auxiliary causal and contrastive terms. Such hybrid architectures enable the model to perform bidirectional latent reasoning, synthesizing contextually coherent behaviors by leveraging both local sequential context and global world knowledge encoded within the pre-trained backbone, ultimately leading to superior performance in cold-start and cross-domain scenarios.

4.1.3 Modeling Higher Order Dependencies and Transition Patterns

Traditional sequential recommendation approaches often rely on first-order Markov assumptions, which fail to capture complex, long-range dependencies inherent in user behavior [28]. To address this limitation, recent methodologies employ Transformer-based architectures equipped with self-attention mechanisms to model higher-order interactions across entire interaction histories. By converting event sequences into composite tokens, these models effectively encode rich, structured information into single vector representations, allowing for the simultaneous processing of thousands of events while preserving intricate temporal dynamics that simple transition matrices overlook [12].

Beyond standard autoregressive modeling, advanced frameworks incorporate graph-based struc-

tures to explicitly represent transition patterns and causal pathways within behavioral data. Weighted directed graphs are constructed to map adjacent events, where edge weights reflect transition frequencies, thereby distinguishing between behavior sequence-originated paths and spurious correlations. This structural approach facilitates the identification of missing intermediate behaviors and smooths semantic transitions by leveraging both left and right contexts, ensuring that the model accurately captures the true generative roles of specific interaction paths rather than treating all historical data uniformly.

Furthermore, emerging techniques integrate foundation models to handle periodic behaviors and uncertainties through sophisticated pre-training objectives, such as next transaction prediction augmented with learnable profile tokens. These strategies often utilize curriculum learning to progress from simpler to more complex tasks, enhancing the model’s ability to generalize across anchor and tail behaviors. By combining bidirectional implicit behavior reasoning with rigorous probabilistic formulations, modern systems achieve robust scalability and improved accuracy in predicting future interactions, effectively bridging the gap between raw sequential data and high-level user intent.

4.2 Constrained and Lookahead Decoding Mechanisms

4.2.1 Lookahead Heuristics and Future Attribute Planning

Lookahead heuristics in generative recommendation shift the paradigm from myopic token prediction to trajectory-aware planning. Unlike standard teacher-forcing approaches that optimize for immediate next-token accuracy, these methods estimate the value of future states to guide current candidate selection [29]. By integrating an approximation of future constraint satisfaction or attribute realization into the decoding score, the model can identify tokens that lie on optimal long-term trajectories. This mechanism effectively mitigates exposure bias and prevents the model from converging to locally optimal but globally suboptimal sequences, which is critical for complex multi-step reasoning tasks.

Future attribute planning extends this concept by explicitly modeling the probability of desired characteristics manifesting in the completed se-

$$q(s_T, y_T) := \begin{cases} 0 & \text{if } y_T = \text{EOS} \\ -\infty & \text{if } y_T \neq \text{EOS} \end{cases}.$$

Figure 27: Chart from ‘Autoregressive Language Models are Secretly Energy-Based Models: Insights into the Lookahead Capabilities of Next-Token Prediction’ (arXiv:2512.15605)

quence rather than the current step. Techniques such as discriminative guidance allow the system to evaluate whether a partial generation will eventually satisfy specific constraints, such as item categories or user intent profiles. This modular approach enables the composition of diverse attributes, including stylistic constraints and semantic topics, without requiring retraining of the underlying foundation model. The planner acts as a lookahead filter, reweighting candidate tokens based on their potential to fulfill these distant objectives, thereby aligning the generation process with high-level user goals.

The integration of lookahead strategies with beam search further enhances planning capabilities by maintaining multiple hypothesis paths simultaneously. Instead of greedily selecting the single highest-probability token, the system preserves a set of top candidates, allowing it to explore branching futures and recover from early missteps. This is particularly advantageous in sparse data regimes where sequential patterns are less distinct, as the lookahead component provides a robust signal derived from attribute consistency rather than solely relying on historical co-occurrence statistics. Consequently, this synthesis of heuristic search and attribute conditioning transforms recommendation from statistical pattern matching into a knowledge-grounded, forward-looking decision process.

4.2.2 Repetition Prevention and Diversity Enhancement

Repetition prevention in generative recommendation systems primarily addresses the tendency of autoregressive models to converge on high-probability tokens, resulting in monotonous user interaction sequences. Standard decoding strategies often fail to capture the stochastic nature of human behavior, leading to collapsed diversity where models repeatedly suggest popular items while ignoring long-tail preferences. To mitigate

this, recent methodologies integrate constraint-based decoding mechanisms that dynamically prune candidate sets containing previously interacted items or semantically redundant clauses. These approaches ensure that generated sequences adhere to logical consistency while strictly enforcing non-repetition constraints at the token level, thereby preserving the structural integrity of user behavior trajectories without compromising prediction accuracy.

Diversity enhancement extends beyond simple exclusion logic by actively promoting the selection of novel and varied items through specialized objective functions and sampling protocols. Techniques such as distinct n-gram metrics and coverage-based losses are employed during training to penalize homogeneous outputs and encourage the exploration of under-represented item categories. Furthermore, advanced frameworks utilize grouping mechanisms that cluster candidate predictions based on semantic features, forcing the model to select high-scoring representatives from diverse groups rather than concentrating probability mass on a narrow subset. This strategic distribution of attention allows the system to maintain high recall rates for rare events and improves the overall robustness of the model against popularity bias inherent in historical interaction data.

The synergy between repetition prevention and diversity enhancement is critical for achieving balanced performance in dynamic environments where user interests evolve rapidly. By combining token-level weighting mechanisms with multi-task learning objectives, models can simultaneously optimize for immediate prediction accuracy and long-term behavioral variety. Empirical analyses indicate that disabling these diversity-promoting components leads to significant performance degradation, particularly in cold-start scenarios and tasks requiring the identification of low-frequency behaviors. Consequently, effective system design must carefully calibrate the trade-off between exploration and exploitation, ensuring that the generative process remains both faithful to user history and sufficiently innovative to facilitate discovery.

4.2.3 Validating Generated Identifiers via Index Structures

Validating generated identifiers necessitates rigorous alignment between the model's output space and the underlying retrieval index to ensure se-

mantic consistency. Traditional approaches often rely on exact string matching or direct ID lookup, which can fail when large language models produce hallucinated or syntactically valid but non-existent identifiers. To mitigate this, recent frameworks integrate index structures directly into the verification loop, treating the index not merely as a storage mechanism but as a constraint solver. This process involves mapping generated token sequences back to discrete item spaces, where invalid mappings are immediately flagged, preventing the propagation of errors in downstream recommendation or retrieval tasks.

Advanced validation strategies employ hierarchical index structures, such as tree-based quantization or inverted files, to verify the plausibility of generated identifiers at multiple granularity levels. By decomposing complex identifiers into semantic sub-tokens, the system can validate partial matches against the index before committing to a full sequence generation. This multi-stage verification reduces the computational overhead associated with exhaustive search while maintaining high precision. Furthermore, incorporating relaxed sampling verification allows the system to accept draft sequences that fall within a probabilistic neighborhood of valid index entries, thereby balancing strict adherence to the catalog with the generative flexibility required for novel user intents.

The integration of index-aware validation also addresses the issue of distributional shift between training data and inference-time generations. By enforcing constraints derived from the index topology, models are guided to explore only feasible regions of the item space, effectively reducing the occurrence of "glitch tokens" or under-trained representations. This structural alignment ensures that the generated identifiers preserve both collaborative filtering signals and semantic attributes encoded within the index. Consequently, the validation step transforms from a passive post-processing filter into an active component of the generation architecture, significantly enhancing the reliability and scalability of identifier-based systems in large-scale deployment scenarios.

4.3 Scaling Laws and Efficient Architecture Design

4.3.1 Compute Optimal Configurations for Behavioral Models

Determining compute-optimal configurations for behavioral models requires navigating a distinct scaling regime that diverges from standard language modeling heuristics. Unlike text-based large language models where data and parameter counts often scale symmetrically, behavioral models exhibit a strong data-heavy bias at limited compute budgets, necessitating a significantly higher ratio of data tokens to parameters. As total compute increases, the optimal configuration gradually shifts toward the language-model regime, yet the specific trajectory remains contingent on the target deployment metric, such as Hit Ratio or Normalized Discounted Cumulative Gain.

The allocation of computational resources between model depth and width, as well as the selection of batch sizes, follows a metric-dependent recipe rather than a universal law. Empirical analysis suggests that critical batch sizes vary according to the specific evaluation objective, influencing the efficiency of gradient updates during the two-stage training process. Furthermore, the utility of negative sampling strategies changes dynamically; the optimal number of negatives required for convergence differs before and after freezing specific model components, indicating that static hyperparameter settings often lead to sub-optimal performance in behavior prediction tasks.

Consequently, achieving robust recommendation performance involves a sequential decision process that prioritizes the deployment metric before configuring architectural hyperparameters. Practitioners must first select an appropriate event embedder and determine batch sizes based on the trade-off between data efficiency and throughput. Subsequently, compute should be allocated between model dimensions and data volume by adhering to scaling laws derived from extensive parameter sweeps. This systematic approach ensures that the behavioral model stack maximizes predictive accuracy while adhering to strict computational constraints, effectively balancing the competing demands of sequence modeling fidelity and inference latency.

4.3.2 Token Compression and Context Window Optimization

Token compression techniques address the computational bottlenecks associated with expanding context windows in large language models by reducing sequence length while preserving semantic fidelity. Methods such as Residual Quantized Variational Autoencoders progressively decompose continuous vectors into hierarchical discrete codes, enabling efficient representation of complex behavioral data. Furthermore, learnable interest anchor groups utilize cross-attention mechanisms to compress extensive user interaction histories into compact representations, capturing diverse interests without retaining every raw token. These approaches effectively mitigate the quadratic complexity of self-attention, allowing models to process longer temporal horizons within fixed memory constraints.

Context window optimization extends beyond simple truncation by employing dynamic strategies to prioritize salient information during inference. Speculative decoding accelerates generation by drafting multiple future tokens via a lightweight model and verifying them in parallel, significantly reducing the number of expensive large model calls [27]. Complementary to this, token-level weighting mechanisms dynamically assign decreasing importance to later tokens in behavior sequences, acknowledging that recent interactions often exert less influence on specific prediction targets compared to earlier contextual cues. This selective focus ensures that the model allocates computational resources to the most impactful segments of the input sequence, enhancing both efficiency and predictive accuracy.

The integration of these compression and optimization strategies facilitates robust performance in retrieval-augmented scenarios where context limits are critical. By summarizing retrieved documents or behavior logs, systems can achieve substantial token reduction rates while maintaining, or even improving, downstream task accuracy compared to using full uncompressed contexts. This balance is crucial for deploying models in resource-constrained environments, as it allows for the internalization of item knowledge and personalized patterns without necessitating frequent system retraining. Ultimately, these advancements enable scalable architectures that natively handle structured events as composite tokens, bridging

the gap between architectural rigidity and the demand for extensive contextual understanding.

4.3.3 Unified Tokenization for Multi-Attribute Events

Unified tokenization for multi-attribute events addresses the semantic fragmentation inherent in standard sub-word tokenizers by treating each complex interaction as a single, indivisible composite unit [12]. Unlike traditional approaches that serialize events into disjointed text sequences, this paradigm constructs tokens via the Cartesian product of contextual and counterpart features, such as payment channels, amounts, and merchant categories. This structural integrity ensures that the model processes the event holistically, preserving critical correlations between attributes that are often lost when splitting identifiers into independent sub-tokens.

The embedding mechanism for these unified tokens integrates textual semantics, positional information, and specific entity identifiers through element-wise addition. For a given token within a sequence, the final representation combines the textual embedding of the attribute string, a positional vector capturing temporal order, and a dedicated ID embedding unique to the user or item involved [30]. Crucially, while item tokens carry item-specific ID embeddings and user tokens carry user-specific ones, all components share the same underlying composite identity, allowing the model to distinguish between different instances of similar attribute combinations without sacrificing semantic coherence.

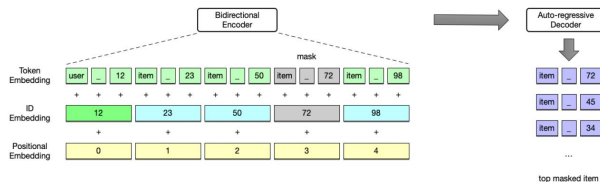


Figure 28: Chart from 'GenRec: Generative Sequential Recommendation with Large Language Models' (arXiv:2407.21191)

This approach significantly enhances the model's ability to perform beyond-sequence reasoning and maintain behavioral continuity across dense representation spaces. By avoiding the local optima issues associated with token-sequence identifiers, unified tokenization enables more efficient inference and robust generalization across diverse behavioral categories. The resulting repre-

sentations facilitate accurate next-token prediction by aligning the input granularity with the intrinsic structure of multi-dimensional events, thereby reducing noise from invalid token alignments and improving the fidelity of user behavior modeling in complex, multi-attribute domains.

4.4 Interpretability and Latent Space Analysis

4.4.1 Probing Latent World Models in Transformers

Recent advancements in probing latent world models within Transformers have shifted focus from mere pattern recognition to understanding the internal representation of global knowledge and temporal dynamics. By leveraging bidirectional encoder representations, these models synthesize local behavioral contexts with pre-trained semantic backbones to perform latent reasoning. This dual capability allows the architecture to infer missing events by reasoning both forward from preceding contexts and backward from subsequent observations, effectively constructing a coherent latent world model that maintains temporal consistency across sparse data sequences.

The efficacy of these latent models is particularly evident in their exceptional sample efficiency and generalization to tail behaviors. Empirical analyses demonstrate that integrating rich, structured event descriptions into the Transformer backbone enables the model to capture long-range dependencies that traditional domain-specific architectures often miss. For rare events occurring with less than one percent frequency, this approach yields substantial performance gains, significantly outperforming standard fine-tuning methods. The latent space thus serves as a robust repository of world knowledge, allowing the model to achieve high accuracy even in few-shot scenarios where training data is extremely limited.

Furthermore, theoretical investigations suggest that the auto-regressive next-token prediction paradigm employed by these Transformers functions as a universal function learner, mathematically equivalent to full-item vocabulary maximum likelihood estimation. This equivalence provides a rigorous foundation for the model's ability to encode complex feature interactions through stacked self-attention layers. As the model depth increases, these layers progressively capture higher-order dependencies, transforming raw input sequences into

implicit representations that align closely with the underlying generative processes of the observed world, thereby validating the Transformer’s role as a powerful engine for latent world modeling.

4.4.2 Disentangling Semantic and Domain Specific Features

Disentangling semantic and domain-specific features is critical for mitigating representation collapse in sequential modeling, particularly when addressing long-tail items and sparse interaction data. Traditional approaches often rely on one-hot encodings or purely collaborative filters that treat events as unrelated IDs, failing to capture inherent associations between activities, such as the link between opening a refrigerator and preheating an oven. To overcome this, recent frameworks leverage Large Language Models (LLMs) to inject intrinsic semantic knowledge, generating behavior embeddings that serve as semantic anchors in the latent space. This allows tail behaviors to cluster around meaningful prototypes, ensuring that low-frequency actions are represented through their semantic proximity to common patterns rather than being lost in noise.

Advanced architectures employ dual-view modeling and attribute-driven learning to explicitly separate content semantics from domain-specific interaction patterns. By utilizing techniques such as cross-expert covariance loss or unified autoencoders, these models decompose concatenated semantic embeddings into distinct latent dimensions, preventing the entanglement of item attributes with user preference signals. For instance, some methods introduce composite tokenization where multi-attribute events are treated as coherent units, enabling the model to learn bidirectional relationships via masked prediction tasks. This separation ensures that the learned representations capture product attributes and usage contexts that remain inaccessible to purely collaborative methods, thereby enhancing the robustness of the feature foundation against distributional shifts.

Furthermore, disentangled representations facilitate superior cross-domain adaptability and generalization by isolating transferable semantic knowledge from domain-confounded biases. Unlike methods dependent on auxiliary data that increase training complexity, disentangled frameworks distill nuanced domain knowledge into semantically grounded item representations without compromising anchor performance. The resulting em-

beddings act as powerful, memory-efficient categorical features that complement continuous vector spaces, allowing models to identify high-level patterns and relationships across diverse datasets. This orthogonality between semantic and domain-specific paradigms yields compounding gains, effectively addressing challenges such as popularity and duration bias while maintaining high fidelity in next-item prediction tasks across complex, multi-domain environments.

4.4.3 Linear Representation Hypothesis in Recommendation

The Linear Representation Hypothesis in recommendation posits that user preferences and item attributes can be effectively modeled as linear combinations within a high-dimensional latent space [31]. This assumption underpins many collaborative filtering and sequential models, where dot-product interactions between user and item embeddings serve as the primary mechanism for scoring relevance. By treating semantic concepts and behavioral patterns as additive vectors, these systems leverage geometric properties to infer unseen interactions, facilitating efficient retrieval and generalization across sparse data distributions without requiring complex non-linear transformation layers for every inference step.

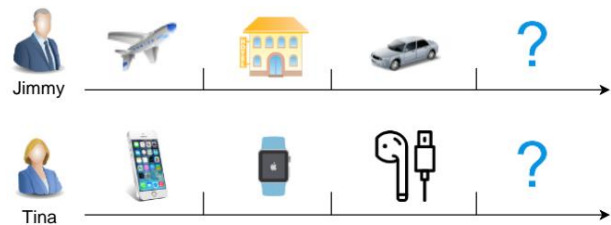


Figure 29: Chart from 'Sequential Recommender Systems: Challenges, Progress and Prospects' (arXiv:2001.04830)

Recent advancements integrating large language models into recommender systems have re-examined this hypothesis through the lens of semantic alignment [3]. Studies suggest that pre-trained language representations inherently encode relational knowledge linearly, allowing for the direct projection of textual item descriptions into collaborative filtering spaces. Techniques such as linear probing and adapter layers are employed to map frozen language embeddings to task-specific recommendation objectives, preserving the linear structure while adapting to domain-specific signals. This approach mitigates the risk

of representation collapse and enables the transfer of world knowledge from language corpora to recommendation tasks with minimal parameter overhead.

However, the validity of strict linearity faces challenges when modeling complex, higher-order user interests that may require non-linear disentanglement. While linear representations offer computational efficiency and interpretability, they may struggle to capture nuanced contextual dependencies present in dynamic interaction sequences. Consequently, contemporary research explores hybrid architectures that maintain a linear backbone for scalability while introducing lightweight non-linear modules to refine latent representations. This balance aims to retain the theoretical benefits of the linear hypothesis while enhancing expressive power to address homogeneity and amplification biases often observed in purely linear generative recommendation frameworks.

5 Future Directions

Despite the significant advancements in generative recommendation systems driven by Large Language Models, several critical limitations persist that hinder their widespread deployment and reliability. A primary challenge remains the susceptibility of these models to hallucinations, where generated item identifiers or recommendations do not correspond to valid entities within the actual catalog, undermining user trust. Furthermore, while current architectures successfully integrate multimodal data and collaborative signals, they often struggle with computational efficiency and latency, particularly when performing autoregressive generation over massive item spaces. The alignment between collaborative filtering objectives and language modeling goals is also not fully resolved, leading to potential conflicts where semantic richness compromises the precision of behavioral pattern recognition. Additionally, existing systems frequently lack robust mechanisms for long-term user state tracking, limiting their ability to maintain coherent personalization across extended, multi-turn interactions without explicit re-prompting.

Future research must prioritize the development of rigorous grounding mechanisms and constrained decoding strategies that strictly enforce factual validity during the generation process. This involves creating hybrid architectures that

seamlessly combine the probabilistic flexibility of language models with deterministic retrieval constraints, ensuring every generated token maps to a verified item in the database. Concurrently, there is an urgent need to optimize inference efficiency through advanced tokenization schemes and speculative decoding techniques that reduce the computational overhead of generating long semantic ID sequences. Another promising direction lies in the evolution of autonomous recommendation agents capable of proactive, multi-step planning. Instead of merely reacting to immediate queries, future systems should leverage symbolic reasoning traces to anticipate user needs, actively seek clarifying information, and dynamically update user profiles based on implicit feedback loops. Research should also explore self-supervised learning paradigms that allow models to continuously adapt to shifting data distributions and emerging trends without catastrophic forgetting, thereby enhancing their resilience in dynamic environments.

Addressing these challenges will fundamentally transform the landscape of digital content discovery, shifting recommender systems from passive filtering tools to active, intelligent partners in the user decision-making journey. By ensuring factual grounding and improving computational scalability, next-generation systems will achieve the reliability required for high-stakes domains such as healthcare, finance, and education, where accuracy is paramount. The integration of autonomous reasoning capabilities will foster deeper user engagement through natural, conversational interactions that mimic human cognitive processes, significantly reducing the friction associated with information overload. Ultimately, these advancements will bridge the gap between raw data and meaningful insight, delivering highly personalized, explainable, and ethically aligned recommendations that adapt fluidly to the complex and evolving nature of human intent.

6 Conclusion

This survey has critically examined the transformative integration of Large Language Models into generative recommendation systems, delineating a shift from traditional discriminative paradigms to unified, semantic-rich frameworks. We explored how semantic identifier tokenization and residual quantization strategies effectively bridge the gap between opaque collaborative signals and continu-

ous language spaces, enabling models to share statistical strength across semantically similar items while mitigating vocabulary explosion. The discussion highlighted the dissolution of boundaries between search and recommendation through joint modeling, where explicit queries and implicit interaction histories are aligned within a single autoregressive backbone. Furthermore, we analyzed the pivotal role of retrieval-augmented generation and multimodal knowledge fusion in grounding predictions, demonstrating how graph-based retrieval and constrained decoding strategies significantly reduce hallucinations while enhancing the system’s ability to reason over heterogeneous data. Finally, the transition toward agent-based architectures was identified as a key frontier, where dynamic context injection and symbolic reasoning traces empower recommenders to engage in multi-turn dialogues and provide narrative-driven explanations that mirror human cognitive processes.

The significance of this comprehensive review lies in its establishment of a novel taxonomy that categorizes the rapidly expanding literature based on generative paradigms, knowledge integration mechanisms, and reasoning architectures. By synthesizing diverse methodologies, this work clarifies the critical trade-offs between the flexibility of generative synthesis and the necessity of factual grounding, offering a structured lens through which researchers can evaluate current advancements. Unlike previous studies that often isolate specific applications or focus solely on discriminative approaches, this survey provides a holistic view of the ecosystem, connecting architectural foundations with practical deployment challenges. It serves as an essential resource for understanding how LLMs redefine core recommender components, from item representation to interactive reasoning, thereby facilitating the development of systems that are not only more accurate but also inherently more interpretable and robust against data sparsity and cold-start scenarios.

Looking forward, the field must address persistent challenges related to computational efficiency, latency in real-time inference, and the rigorous evaluation of generative fidelity beyond standard accuracy metrics. Future research should prioritize the development of lightweight adaptation techniques that allow massive language models to operate within resource-constrained environments without sacrificing reasoning capabilities. Additionally, there is an urgent need for stan-

dardized benchmarks that assess the quality of explanations, the fairness of generated recommendations, and the system’s resilience against adversarial prompts. We call upon the research community to move beyond static prediction tasks and embrace the vision of recommenders as active, reasoning partners capable of dynamic cognitive engagement. By fostering interdisciplinary collaboration between natural language processing, information retrieval, and human-computer interaction, the next generation of recommender systems can achieve a profound alignment with human intent, ultimately transforming digital content discovery into a more intuitive, transparent, and personalized experience.

References

- [1] RecAI Leveraging Large Language Models for Next-Generation Recommender Systems
- [2] Heterogeneous Knowledge Fusion A Novel Approach for Personalized Recommendation via LLM
- [3] The Application of Large Language Models in Recommendation Systems
- [4] Enhancing Job Recommendation through LLM-based Generative Adversarial Networks
- [5] FROM INTERESTS TO INSIGHTS AN LLM APPROACH TO COURSE RECOMMENDATIONS USING NATURAL LANGUAGE-QUERIES
- [6] PALR Personalization Aware LLMs for Recommendation
- [7] Legommenders A Comprehensive Content-Based Recommendation Library with LLM Support
- [8] A Survey on Generative Recommendation Data, Model, and Tasks
- [9] Large Language Models as Data Augmenters for Cold-Start Item Recommendation
- [10] TALKPLAY Multimodal Music Recommendation with Large Language Models
- [11] Fine-grained Semantics Integration for Large Language Model-based Recommendation
- [12] Unified Interaction Foundational Model (UIFM) for Predicting Complex User and System Behavior
- [13] Ethical AI prompt recommendations in large language models using collaborative filtering
- [14] Memory Assisted LLM for Personalized Recommendation System

- [15] GenRec Large Language Model for Generative Recommendation
- [16] Denoising Pre-Training and Customized Prompt Learning for Efficient Multi-Behavior Sequential Recommendation
- [17] Leveraging LLMs and Heterogeneous Knowledge Graphs for Persona-Driven Session-Based Recommendation
- [18] LLMDiRec LLM-Enhanced Intent Diffusion for Sequential Recommendation
- [19] A Survey of Generative Search and Recommendation in the Era of Large Language Models
- [20] Recommendation as Language Processing (RLP) A Unified Pretrain, Personalized Prompt & Predict Paradigm (P5)
- [21] LIBER Lifelong User Behavior Modeling Based on Large Language Models
- [22] Bridge the Domains Large Language Models Enhanced Cross-domain Sequential Recommendation
- [23] E4SRec An Elegant Effective Efficient Extensible Solution of Large Language Models for Sequential Recommendation
- [24] Enhancing Sequential Recommendation with World Knowledge from Large Language Models
- [25] A Survey on Large Language Models for Personalized and Explainable Recommendations
- [26] Generative Conversational Recommender System
- [27] Efficiency Unleashed Inference Acceleration for LLM-based Recommender Systems with Speculative Decoding
- [28] Leveraging Large Language Models for Sequential Recommendation
- [29] Autoregressive Language Models are Secretly Energy-Based Models Insights into the Lookahead Capabilities of Next-Token Prediction
- [30] GenRec Generative Sequential Recommendation with Large Language Models
- [31] Sequential Recommender Systems Challenges, Progress and Prospects